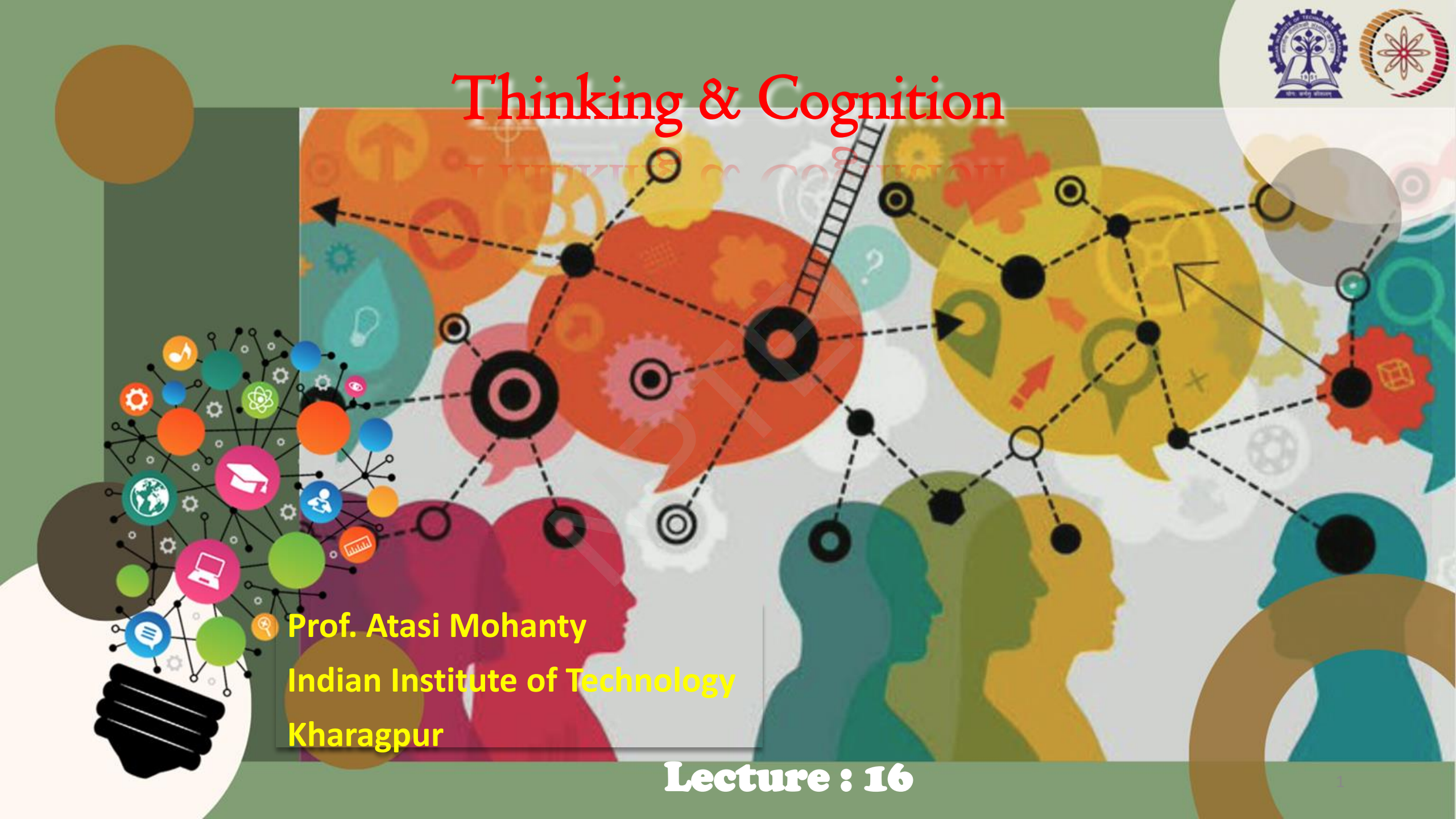


# Thinking & Cognition



**Prof. Atasi Mohanty**  
**Indian Institute of Technology**  
**Kharagpur**

**Lecture : 16**

## • Introduction

- Cognition is a term referring to the mental processes involved in gaining knowledge and comprehension.
- These cognitive processes include thinking, knowing, remembering, judging, and problem-solving.
- These are higher-level functions of the brain and encompass language, imagination, perception, and planning.
- Thinking in psychology is the process of consciously generating and manipulating thoughts and ideas in the mind.
- It allows us to solve problems, learn new information, understand concepts, and process our experiences.
- Thinking involves the entire process of learning, remembering, and organizing mentally to understand the information better and recall it later.
- Primarily there are three main types of thinking in psychology: creative, divergent, and symbolic.

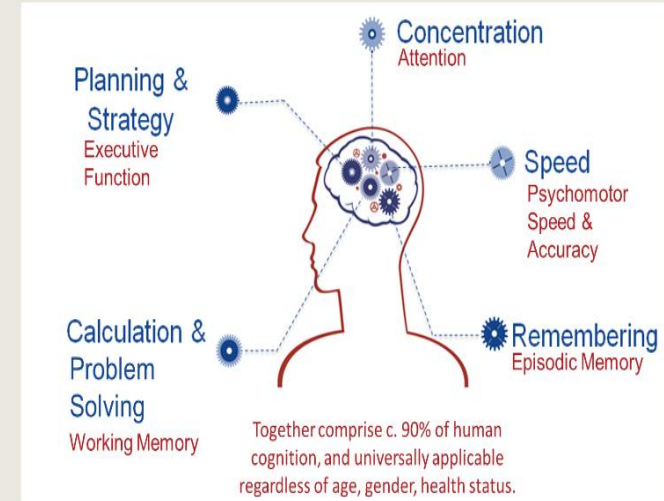




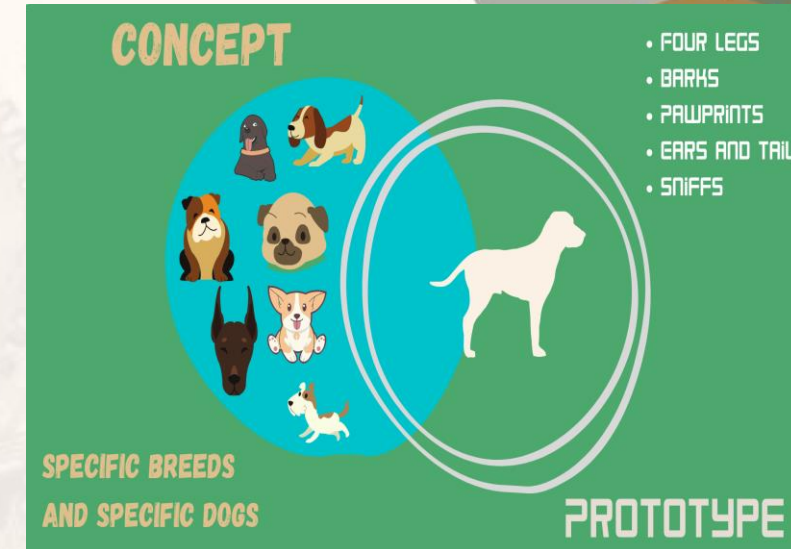


- Creative thinking is the ability to generate innovative, unconventional, or useful ideas.
- Research shows that creativity and intelligence are related to each other somehow ;
- There are also other factors involved in creativity; a person's imagination, environment, and personality can influence their creative thinking abilities.
- When there are many possible answers to a solution, we rely on divergent thinking to help us choose the best solution, like trying to answer or solve an open-ended question.
- Symbolic thinking is the ability to create mental representations of objects, places, events, or people in your mind.
- Young children do this often when they engage in imaginative play; they turn toys and playhouses into symbols of real things.
- **Characteristics of Thinking in Psychology-**
- People with strong thinking skills tend to have several characteristics in common.

#### Five Key Cognitive Domains



- An **active imagination**, a **creative environment**, and an adventurous or **curious personality** can influence our thinking abilities.
- Those who possess expertise in a particular field and **intrinsic motivation** also tend to have more developed thinking skills.
- **Concepts** and **Prototypes** are also important features of Thinking.
- A **concept** is a **mental category of similar objects**, people, or events.
- **Prototypes** are representative items within each concept.
- They are basic mental examples of objects or people, like dogs, doctors, or police officers.
- We compare new information to our **prototype** of the person, place, or thing, so we know how to categorize the new information mentally.
- **Thinking in psychology** is the process of **consciously generating** and **manipulating thoughts** and **ideas** in the mind.



- **Thinking as a Cognitive Process:**
- **Thinking** is the cognitive process, the **use of everything** from **long-term memory to come to** an end or **solution to the problem**.
- Psychologists defined thinking as the **manipulation** (handle) of the **mental representation of information**.
- A mental representation may take the form of a **word, a visual image, a sound, a data** any other sensory modality stored in memory.
- **Thinking** transforms a particular representation of information into a new and different form, allowing us to answer questions, solve problems or reach goals.
- **Thinking** is a **Symbolic process**; the memory representations or symbols for a purpose.
- The mental images or visual representations may be in the **form of** past knowledge, understanding of **signs, symbols, concepts, ideas**, visual **pictures, words, languages, numbers**, and diagrams rather than the overt activity.
- The **systematic arousal of symbols helps organize thinking**.







- Thinking Is A Higher Mental Process-
- It deals with higher mental processes which **examine thinking, knowing and reasoning, problem-solving, decision making**, and understanding the world, communicating our thoughts to others, using our memory and current experience.
- It is only through thinking that **we encode, process, store, retrieve** and **analyze information**.
- Thinking Is A Sub-Vocal Talking-
- It **involves implicit or inner behavior, mental images**, and some slight muscular movements.
- We not just think only in verbal and mathematical symbols but also in **gestures** of the arm, hands, and heads, as well as in a wide **variety of expressive movements**, such as shrug of the shoulders, arrowed eyebrows, a smile, etc. thus the whole organism/biological system is involved in thinking.
- Thinking Is Fulfilling A Need or Problem Solving-

### The process of thinking

The process of thinking is the cognitive mental process which, through analytic and synthetic operations, abstracting and generalizing ones, obtains products under the form of an idea, concept or judgment.

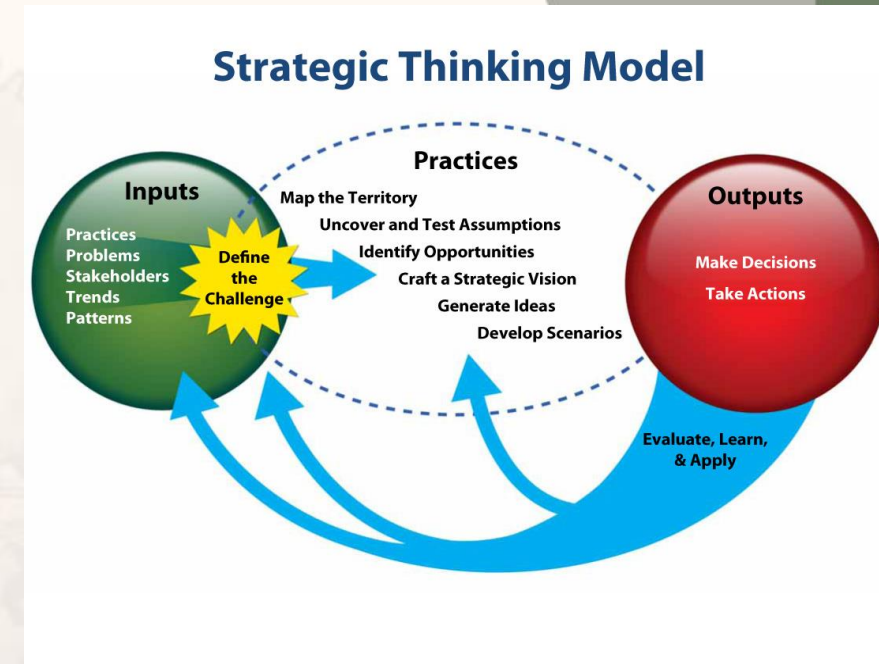


- It is always initiated by a **problem** and ends with its **solution**.
- It is evident when we produce new **meanings, beliefs, and plans** to solve problems or make decisions.
- Some problems are easy to solve while others require great effort, concentration, patience, and thought.
- The thinking goes on to trace out the required **route to reach the goal, mentally removing obstacles**.
- This process helps to **understand the nature of the problem**, identify the needed information to solve it, and then generate a list of **solutions**.
- Thinking can be seen in the **“aha” (I got the answer) experience** when people find the solution to a problem suddenly that had been in their mind for hours or days, later.
- **Elements of Thought-**
  - **Mental images** are **mental pictures of our past experiences**.
  - Whenever we think of certain objects or events **we visualize** them in their absence.
  - **Images can be auditory, olfactory, visual, gustatory, or kinesthetic.**



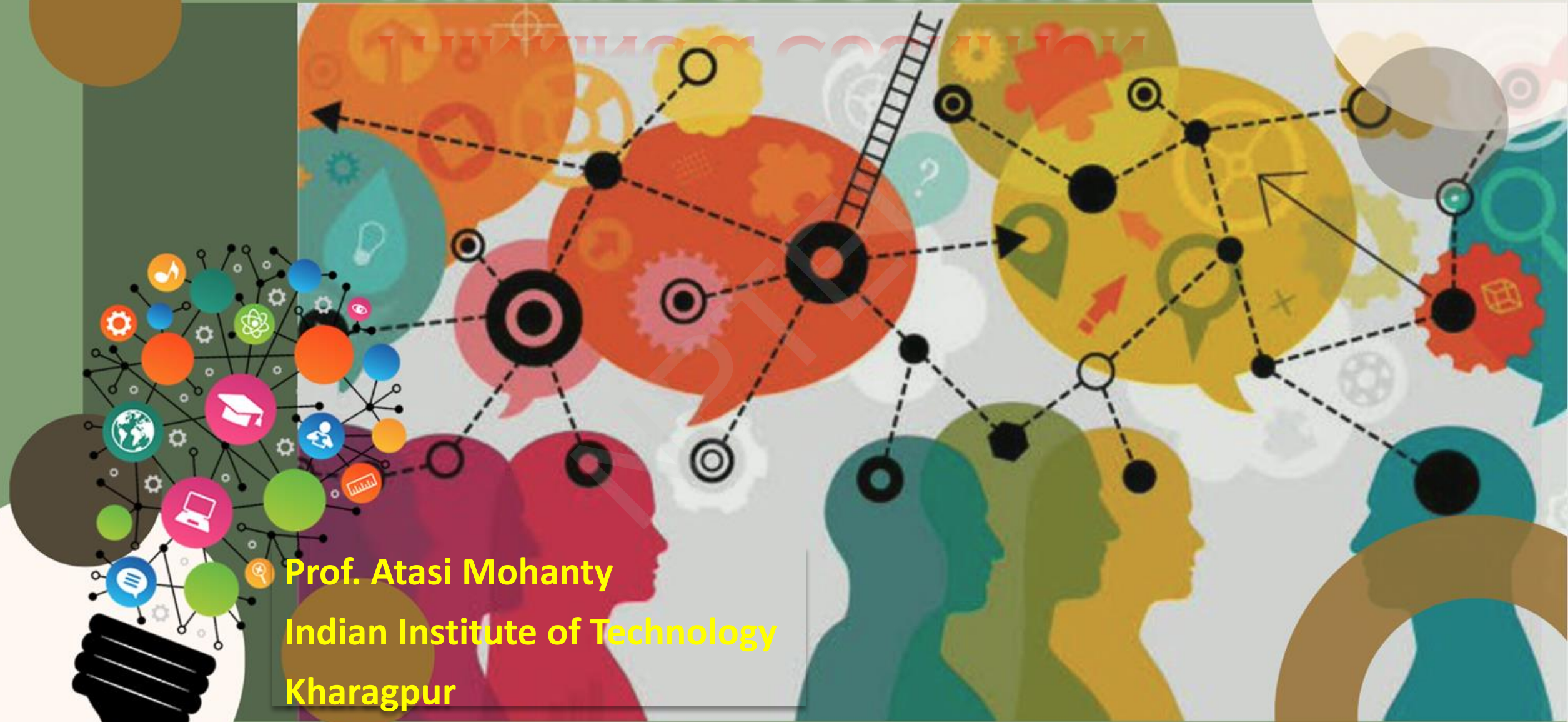


- **Concepts** are the **mental representation** of a class of objects, people, events, and things that share common characteristics. e.g. animals, furniture, vegetables, etc.
- Concepts **help to reduce our mental efforts** and make communication easier with relative ease.
- They **organize complex phenomena into simpler**, easily understandable, and usable categories and help to solve our problems.
- **Propositions**-Thinking **involves languages, percepts**, and symbols to give meaning to our thinking process.
- Propositions describe the **mental manipulation** of these **cognitive skills (languages, mental images, concepts)** for meaningful relationships to what we are thinking.
- Propositions **form mental models** which guide our knowledge structure with objects places and events of the world around us.





# THINKING & COGNITION

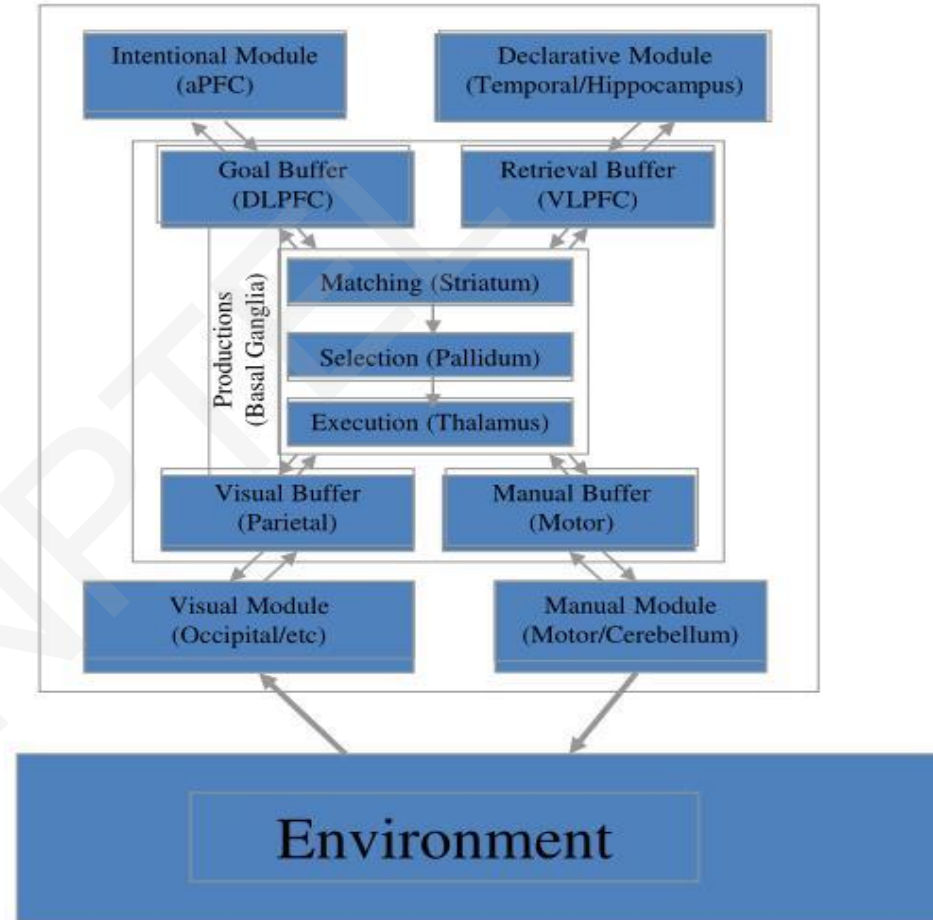


**Prof. Atasi Mohanty**  
**Indian Institute of Technology**  
**Kharagpur**

**Lecture : 17**

# Cognitive Architectures

- Computational representation of invariant cognitive mechanisms
- Behavior selection
  - Production systems
  - Utility – rewards and costs
- Memories
  - Working memory: buffers
  - Long-term: semantic/episodic
  - Activation mechanisms
- Learning
  - Symbolic and statistical
- Human factor limitations
  - Perceptual-motor parameters
- Individual differences
  - Strategies and knowledge
  - Capacity parameters





## • Cognitive Architecture-

- It is a theory about the structures that create a mind in natural or artificial systems.
- It focuses on how these structures work with each other and use the knowledge and skills that are incorporated into the architecture to create and manage intelligent behavior in various complex environments.
- A cognitive architecture alone usually is not able to describe human performance on any specific task, it needs to be provided information about how to carry out that particular task.
- This information is usually based on a thorough task analysis of the target activity that is being modeled.
- A cognitive architecture is also a piece of executable software.
- It is code written by a programmer or several programmers .
- This is a major way in which cognitive architectures are different from the majority of theories in cognitive psychology.

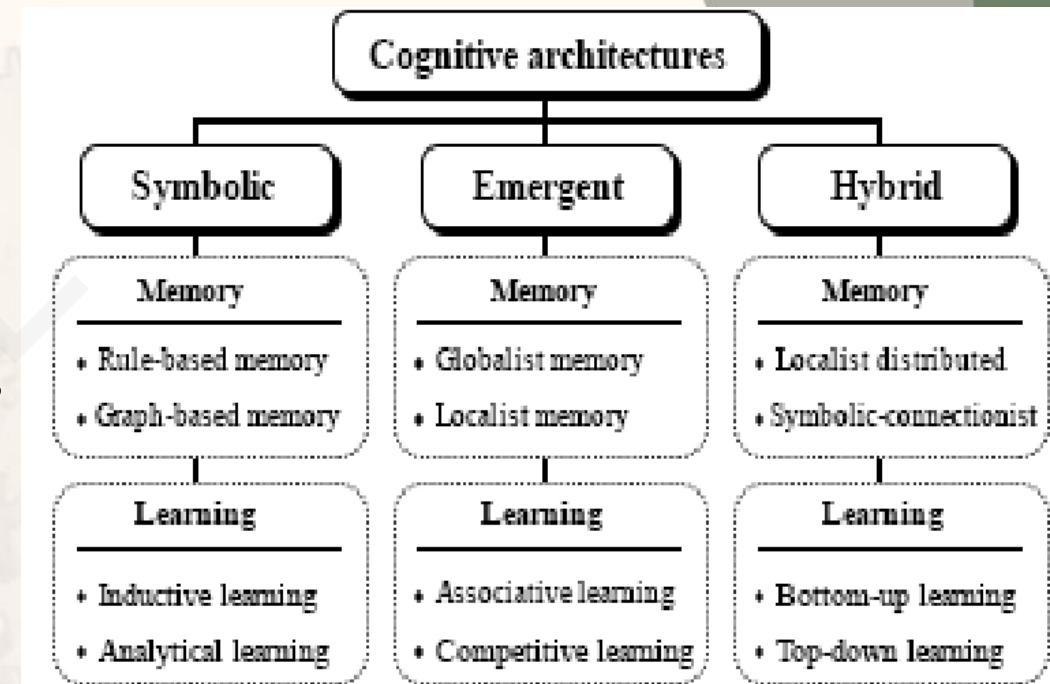
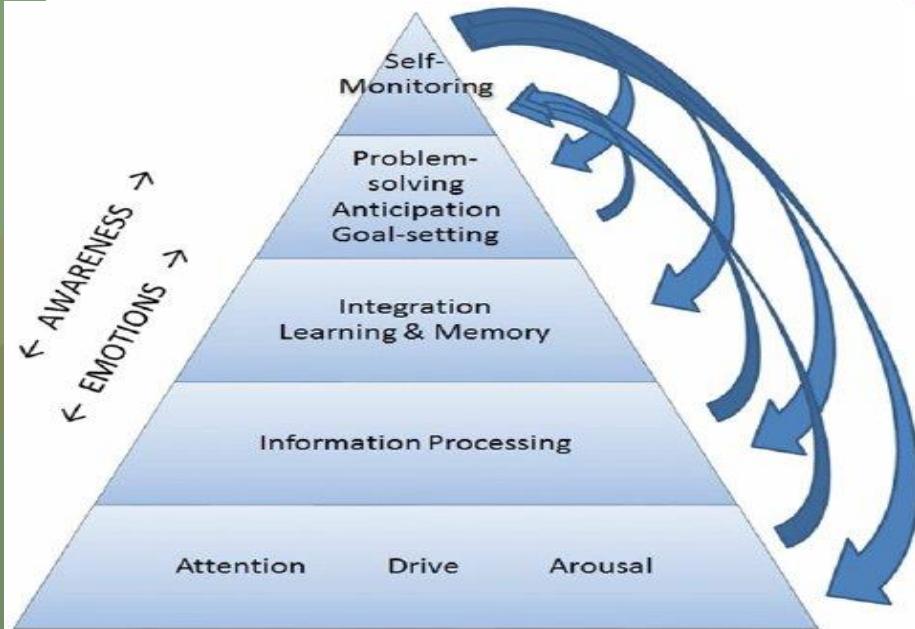
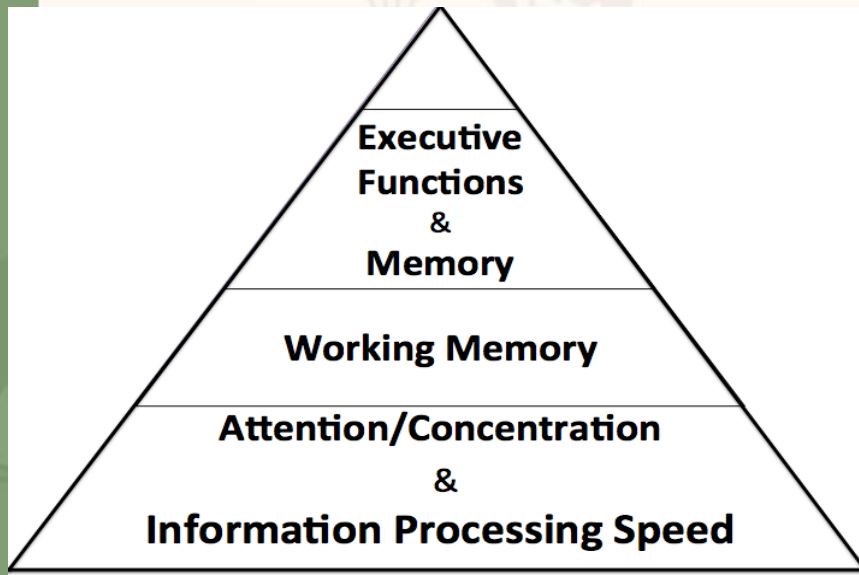
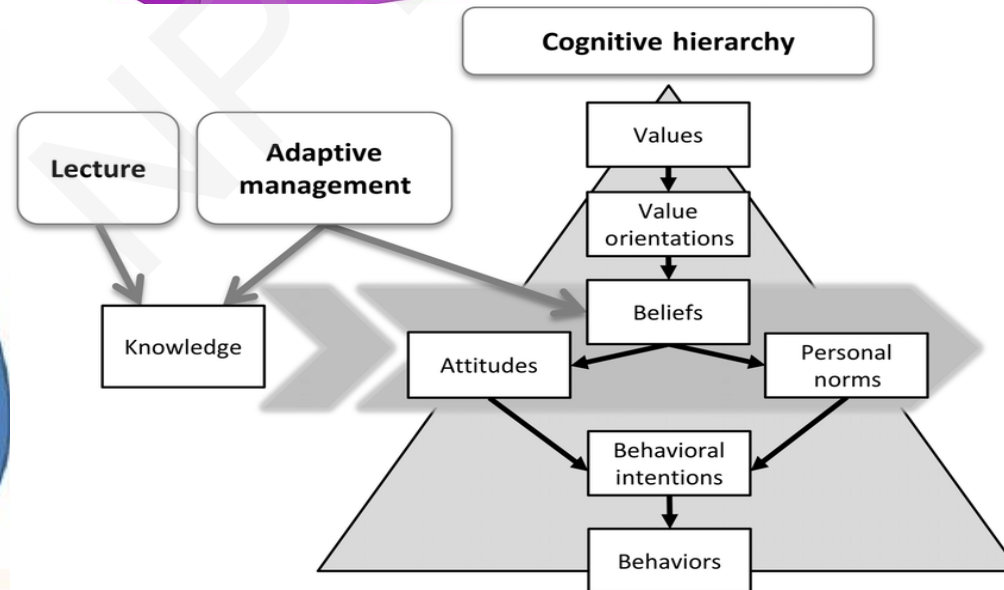
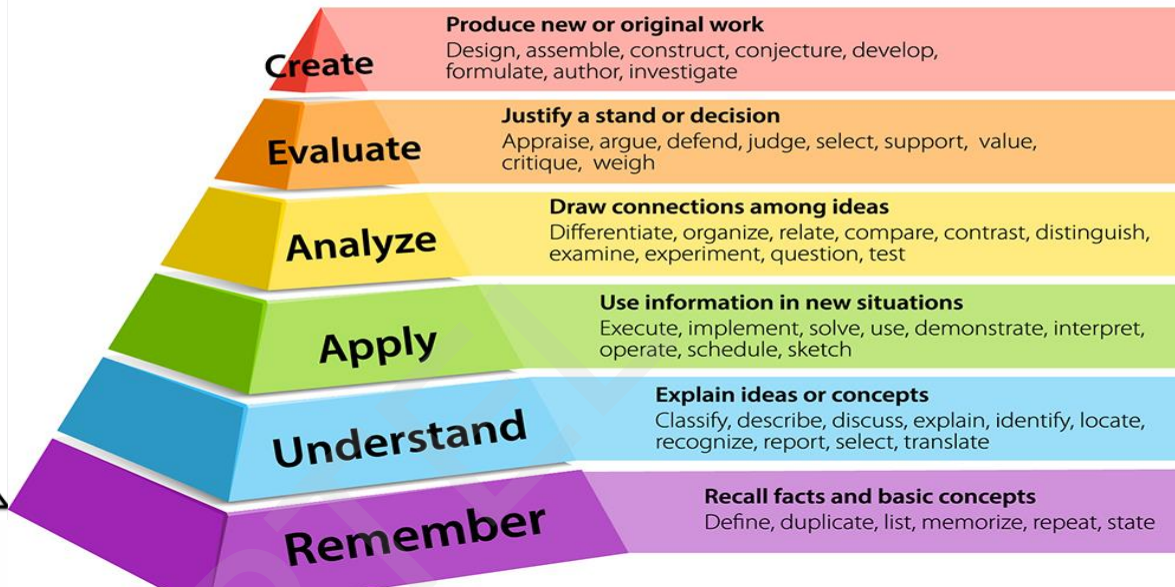


Figure 1: Simplified taxonomy of cognitive architectures

## • Cognitive Hierarchy –



## Bloom's Taxonomy



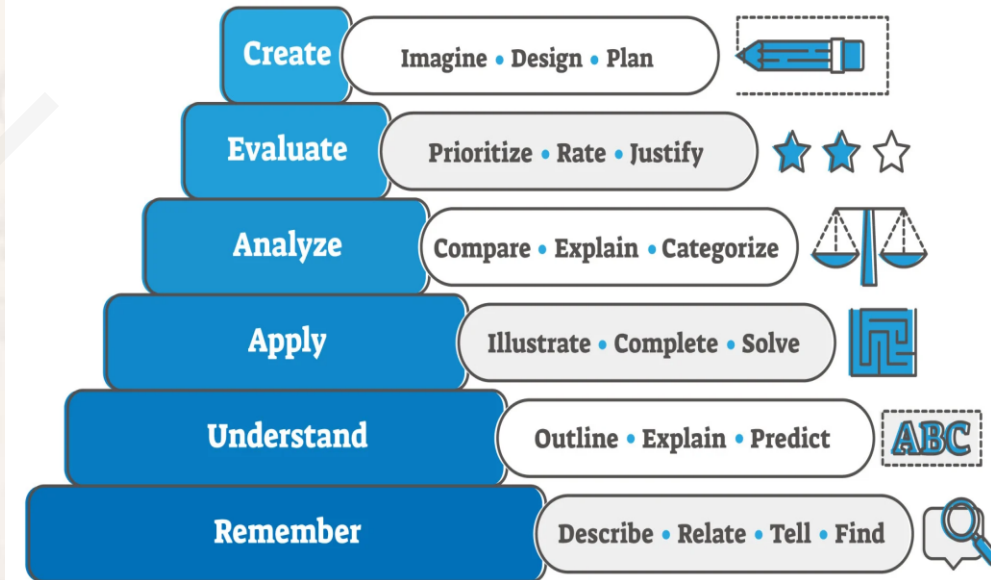




- **Bloom's Taxonomy Of Learning-**

- Bloom's Taxonomy is a system of **hierarchical models** (arranged in a hierarchical rank,) used to categorize learning objectives into varying levels of complexity (Bloom, 1956).
- Bloom's Taxonomy **classifies learning objectives** for students, from recalling facts to producing new and original work.
- It comprises of **three learning domains**: **cognitive**, **affective**, and **psychomotor**.
- Within each domain, learning can take place at a number of levels ranging from simple to complex.
- Bloom's(1971) **Mastery Learning theory** states that students would be able to master subjects when teachers relied upon **suitable learning conditions** and clear learning objectives as guided by Bloom's Taxonomy.
- **The Revised Taxonomy -In 2001**, the original cognitive model was modified by educational psychologists David Krathwol (a co- researcher of Bloom) and Lorin Anderson (a student of Bloom) and published as "**A Taxonomy for Teaching, Learning, and Assessment**".

## BLOOM'S TAXONOMY





- This revised taxonomy emphasizes a more dynamic approach to education; utilizes verbs to describe the active process of learning and does away with the nouns used in the original version (Armstrong, 2001).

| Bloom's Level | Key Verbs (keywords)                                                                                        | Example Learning Objective                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Apply         | calculate, predict, apply, solve, illustrate, use, demonstrate, determine, model, perform, present.         | <i>By the end of this lesson, the student will be able to calculate the kinetic energy of a projectile.</i>                  |
| Understand    | describe, explain, paraphrase, restate, give original examples of, summarize, contrast, interpret, discuss. | <i>By the end of this lesson, the student will be able to describe Newton's three laws of motion to in her/his own words</i> |
| Remember      | list, recite, outline, define, name, match, quote, recall, identify, label, recognize.                      | <i>By the end of this lesson, the student will be able to recite Newton's three laws of motion.</i>                          |



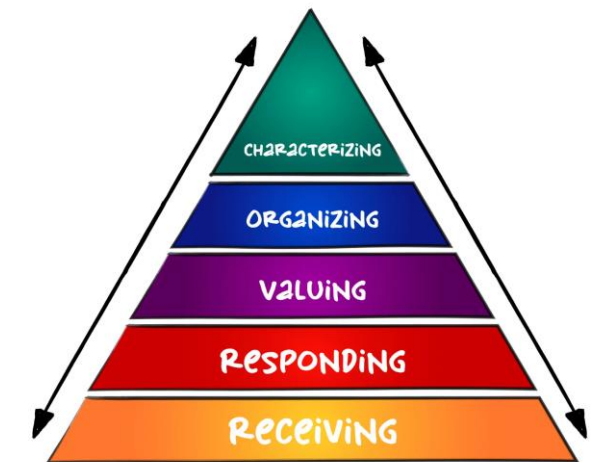


| Bloom's Level | Key Verbs (keywords)                                                                                                              | Example Learning Objective                                                                                                                                                                  |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Create        | design, formulate, build, invent, create, compose, generate, derive, modify, develop.                                             | <i>By the end of this lesson, the student will be able to design an original homework problem dealing with the principle of conservation of energy.</i>                                     |
| Evaluate      | choose, support, relate, determine, defend, judge, grade, compare, contrast, argue, justify, support, convince, select, evaluate. | By the end of this lesson, the student will be able to determine whether using conservation of energy or conservation of momentum would be more appropriate for solving a dynamics problem. |
| Analyze       | classify, break down, categorize, analyze, diagram, illustrate, criticize, simplify, associate.                                   | <i>By the end of this lesson, the student will be able to differentiate between potential and kinetic energy.</i>                                                                           |

- **The Affective Domain (1964)** –(Concerned with feelings and emotion)
- The affective model came as a second handbook (with the first being the cognitive model) and an extension of Bloom's original work (Krathwol et al., 1964).
- This domain focuses on the ways in which we **handle** all things related to **emotions, such as feelings, values, appreciation, enthusiasms, motivations, and attitudes** (Clark, 2015).

## BLOOM'S TAXONOMY

THE AFFECTIVE DOMAIN - EMOTION-BASED



### 1. **Receiving** : basic awareness →

Example : Listening and remembering the names of your classmates when you meet them on the first day of school.

### 2. **Responding** : active participation and reacting to stimuli, with a focus on responding →

Example : Participating in a class discussion.

### 3. **Valuing** : the value that is associated with a particular object or piece of information, ranging from basic acceptance to complex commitment; values are somehow related to prior knowledge and experience →

Example : Valuing diversity and being sensitive to other people's backgrounds and beliefs.





4. **Organizing** : sorting values into priorities and creating a unique value system with an emphasis on comparing and relating previously identified values →

Example : Accepting professional ethical standards.

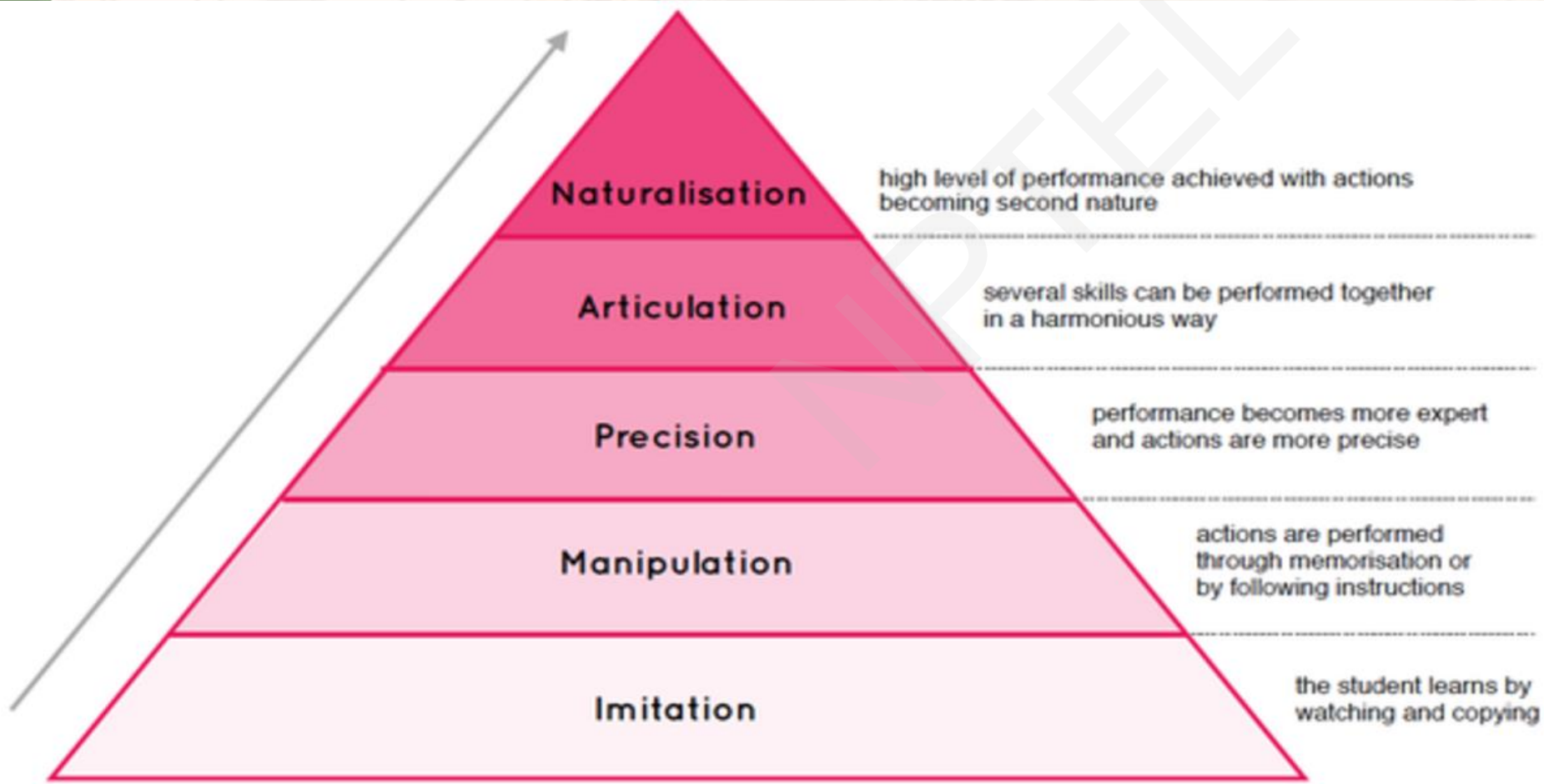
5. **Characterizing** : building abstract knowledge based on knowledge acquired from the four previous tiers; value system is now in full effect and controls the way you behave →

Example : Displaying a professional commitment to ethical standards in the workplace.

## **The Psychomotor Domain (1972)**-Concerned With Skilled Behavior

- The psychomotor model focuses on physical movement, coordination, and anything related to motor skills.
- **Mastery** of these specific skills is marked by speed, precision, and distance.
- These psychomotor skills range from simple tasks, such as washing a car, to more complex tasks, such as operating an intricate piece of technological equipment.
- This model was first published by Robert Armstrong and colleagues in 1970 and included five levels:

- 1) imitation; 2) manipulation; 3) precision; 4) articulation; 5) **naturalization**. These tiers represent different degrees of performing a skill from exposure to mastery.
- Two years later, Anita Harrow (1972) proposed a revised version with six levels: 1) reflex movements; 2) fundamental movements; 3) perceptual abilities; 4) physical abilities; 5) skilled movements; 6) non-discursive communication.





- <https://uwaterloo.ca/centre-for-teaching-excellence/catalogs/tip-sheets/blooms-taxonomy>
- Although knowledge might be the most intuitive block of the cognitive model pyramid, this dimension is actually broken down into four different types of knowledge:
  - **Types of Knowledge-**
  - **Factual knowledge** refers to knowledge of terminology and specific details.
  - **Conceptual knowledge** describes knowledge of categories, principles, theories, and structures.
  - **Procedural knowledge** encompasses all forms of knowledge related to specific skills, algorithms, techniques, and methods.
  - **Metacognitive knowledge** defines knowledge related to thinking — knowledge about cognitive tasks and self-knowledge (“Revised Bloom’s Taxonomy,”).

# THINKING & COGNITION

## Lecture : 18



Prof. Atasi Mohanty  
Indian Institute of Technology  
Kharagpur



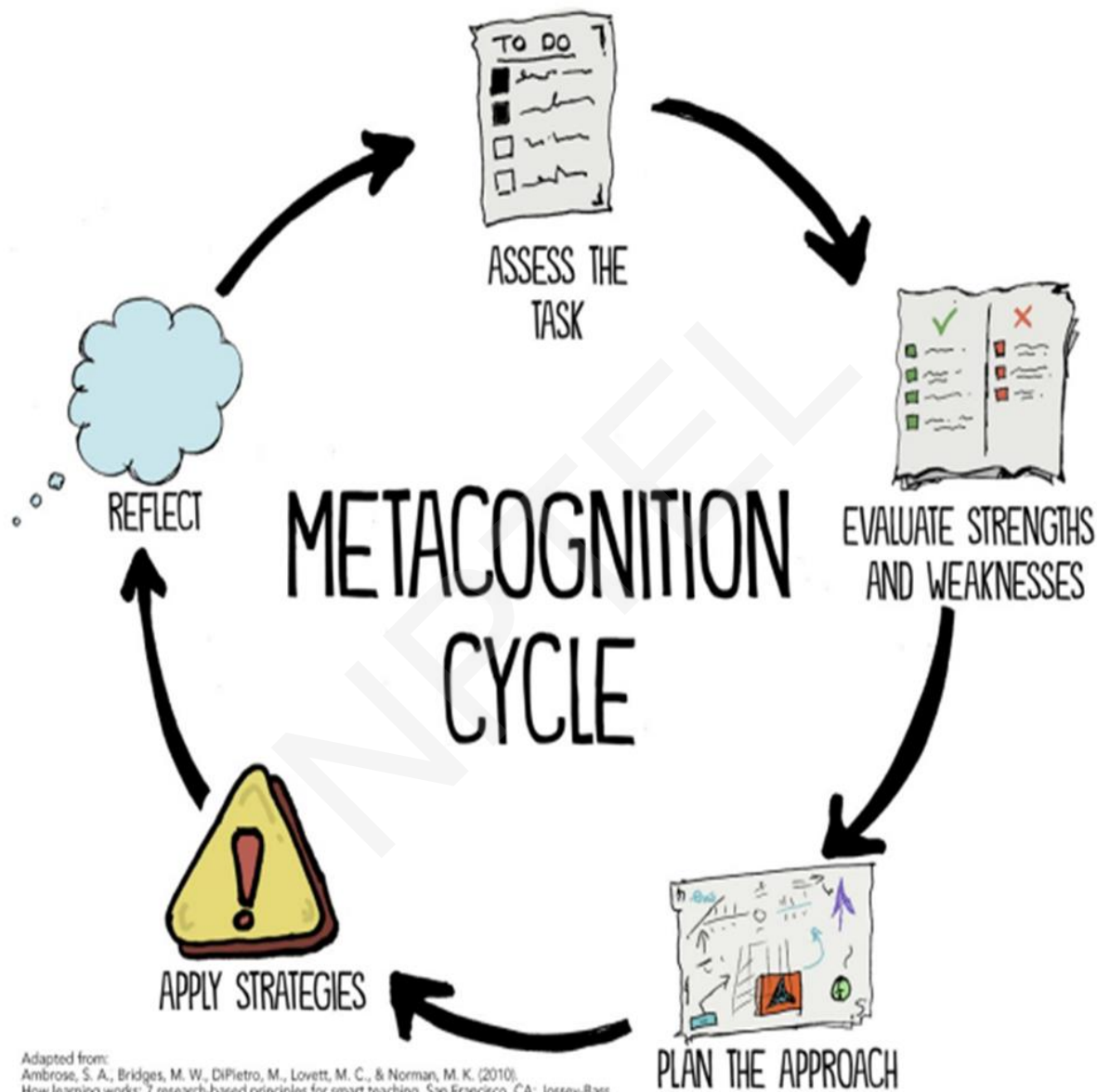
# THE POWER OF METACOGNITION

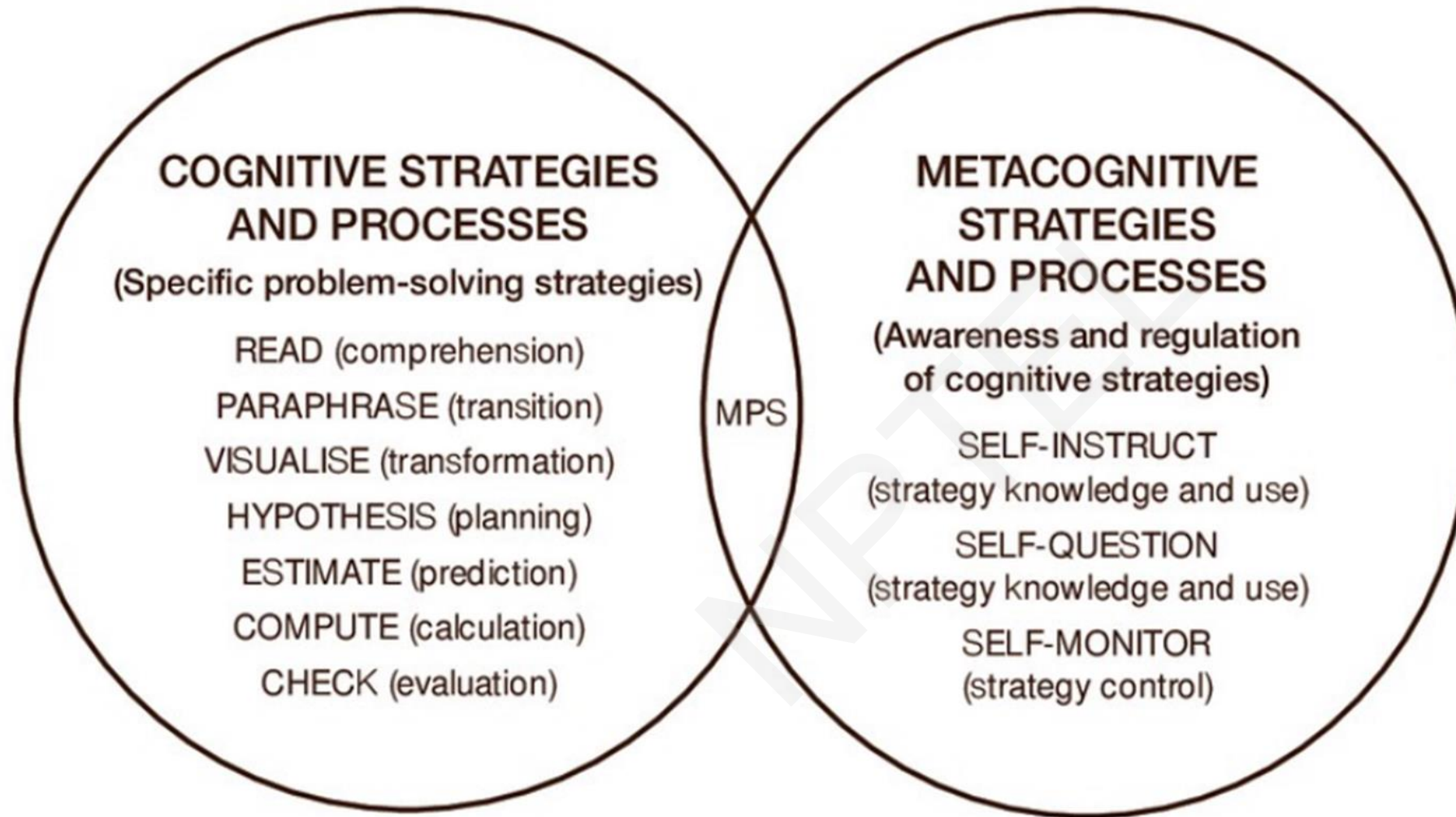
Metacognition is vital for students to thrive in college, in their careers, and in life-long learning. It helps promote autonomy and resiliency. When students improve their metacognitive skills, they are more likely to embrace a Growth Mindset and learn from mistakes.

If we want students to grow into problem-solvers and critical thinkers, we need to help them develop metacognition.



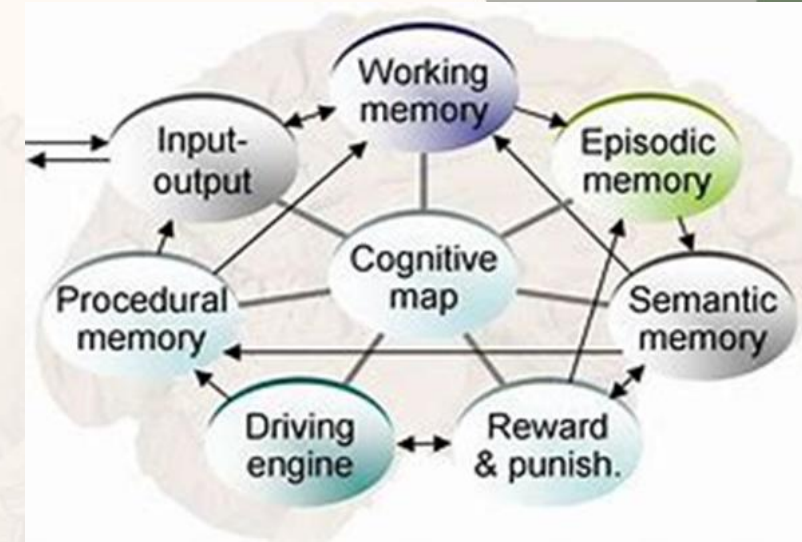
JOHN SPENCER







- **Meta-cognition**
- **Metacognition** is often referred to as “**thinking about thinking**”
- Meta-cognition is a **regulatory system** that helps a person understand and control his or her own **cognitive performance**
- Meta-cognition allows people **to take charge of their own learning**
- Involves awareness of **how we learn**, an **evaluation of our learning needs**, **generating strategies** to meet these needs and then **implementing the strategies**,
- Learners often show an increase in **self-confidence** when they build **meta-cognitive skills**
- For all learners, meta-cognitive knowledge is **crucial for efficient independent learning**, because it fosters **self-reflection**
- **Self-efficacy improves motivation** as well as **learning success**; it fosters forethought

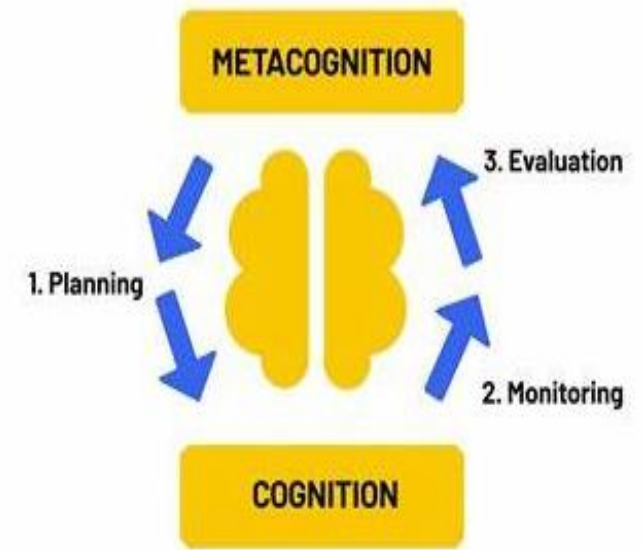


- **Two processes of Metacognition:**

- 1. Knowledge of cognition has three components: **knowledge of the factors** that influence one's own performance;
- knowing different types of strategies to use for learning;
- knowing what strategy to use for a specific learning situation
- 2. **Regulation of cognition involves:** setting goals and planning; monitoring and controlling learning; and evaluating one's own regulation (assessing results and strategies used)

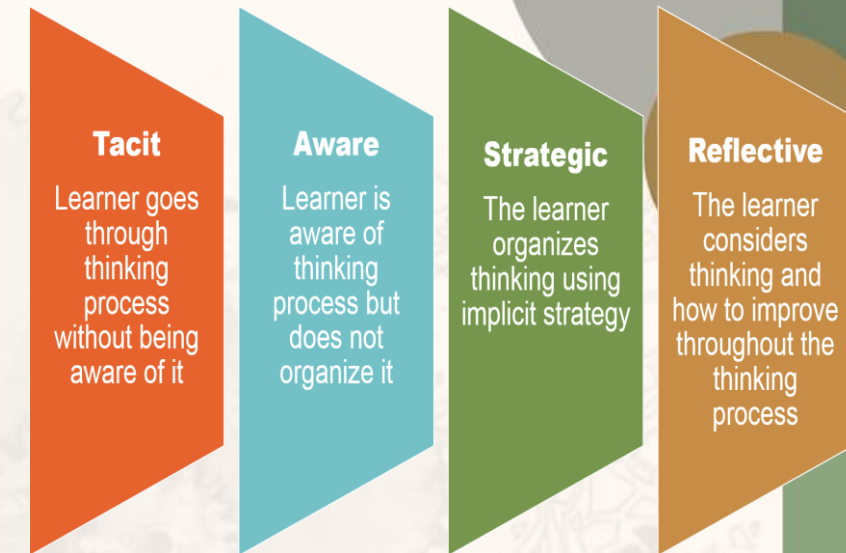
- **Examples of meta-cognition skills**

- 1. **Knowing your limits-** Knowing the limits of our own memory for a particular task and creating a means of external support
- 2. **Self-Monitoring-** Self-monitoring our learning strategy, such as concept mapping, and then adapting the strategy if it isn't effective
- 3. **Modify-** Noticing whether we comprehend something we just read and then modifying our approach if we did not comprehend it.





- **4. Skimming-** Choosing to skim subheadings of unimportant information to get to the information we need
- **5. Rehearsing-** Repeatedly rehearsing a skill in order to gain proficiency (i.e. practice)
- **6. Self-Tests** - Periodically doing self-tests to see how well we learned something
- **Meta-cognitive strategies:** Meta-cognitive strategies facilitate learning how to learn
- **1. Asking Questions.**-allows learners to reflect on our own learning processes and strategies ; allows for enhancing comprehension
- **2. Foster Self-Reflection.**- Critically analyze our own assumptions and how these assumptions have influenced our learning
- **3. Adopt Autonomous Learning.**- Challenge how we learn information; Is it the most efficient and effective way of learning?
- If not try new strategies as a trial and error experiment until we find a strategy that works for us.





- **4. Find a Mentor.** -Many people learn best by interacting with peers who are slightly more advanced; Look for opportunities to observe the proficient use of a skill by a mentor, ask questions and learn from their learning experiences;
- ✓ **5. Find a Group:** Cooperative problem solving can enhance meta-cognitive strategies by discussing possible approaches with group members and learning from each other; can be an excellent way of testing our knowledge on a given subject
- ✓ **6. Think Aloud.-** Reporting our thoughts while performing a difficult task; discovering any potential errors in thinking to address any sabotaging thoughts or language while learning new information
- ✓ **7. Self-Explanation.** -Talking aloud while learning new information can help us improve our comprehension of a difficult subject and bring to light our gaps in retention

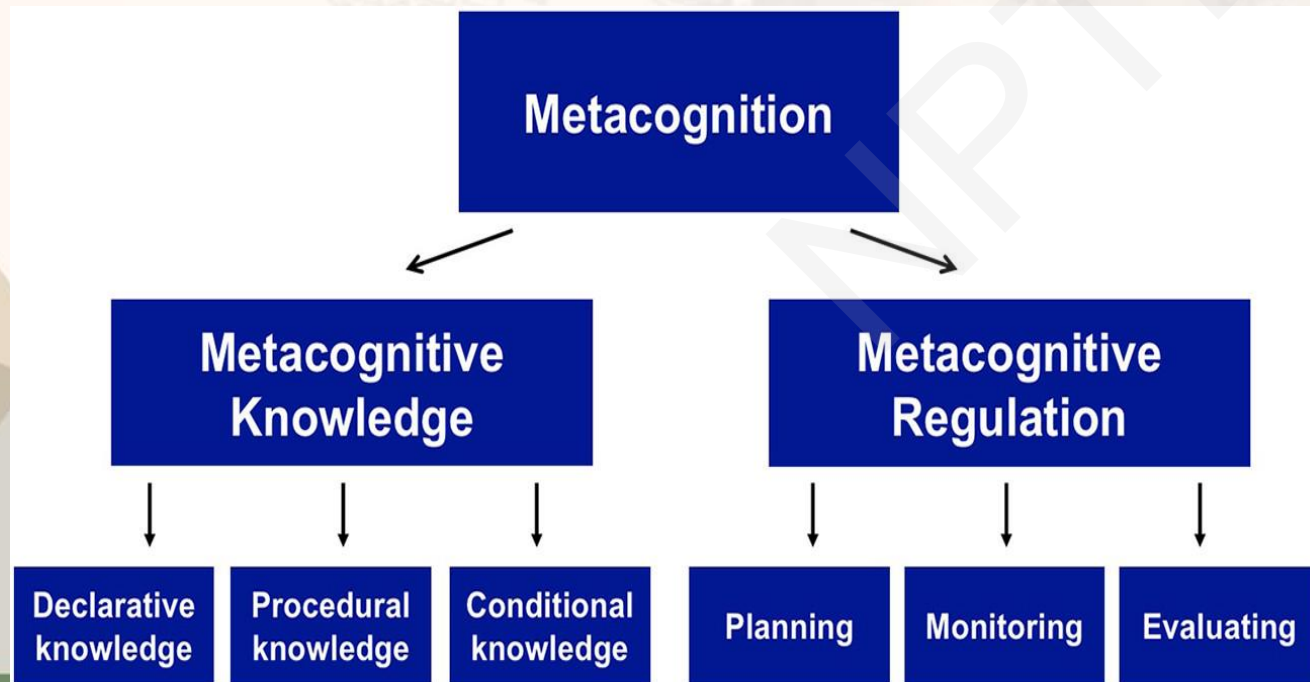
#### Metacognitive Strategies

- Brainstorming
- Webbing
- Task-specific graphic organizers
- Think Aloud Problem Solving-- Self-questioning
- Debate
- Role Play
- Story map
- Story retelling
- Story frame
- Semantic mapping



## ✓ 8. Be ok with Making Errors.-

- ✓ To be given the opportunity to make errors while learning ;
- ✓ **allowing ourselves to make mistakes** as learning is about the process of **getting to know** information, not the arrival at knowing information perfectly
- ✓ Meta-cognitive knowledge **develops with age, experience, and instruction**
- Examples of Cognitive and Meta-cognitive Relationships:
- <https://spencerauthor.com/metacognition/>



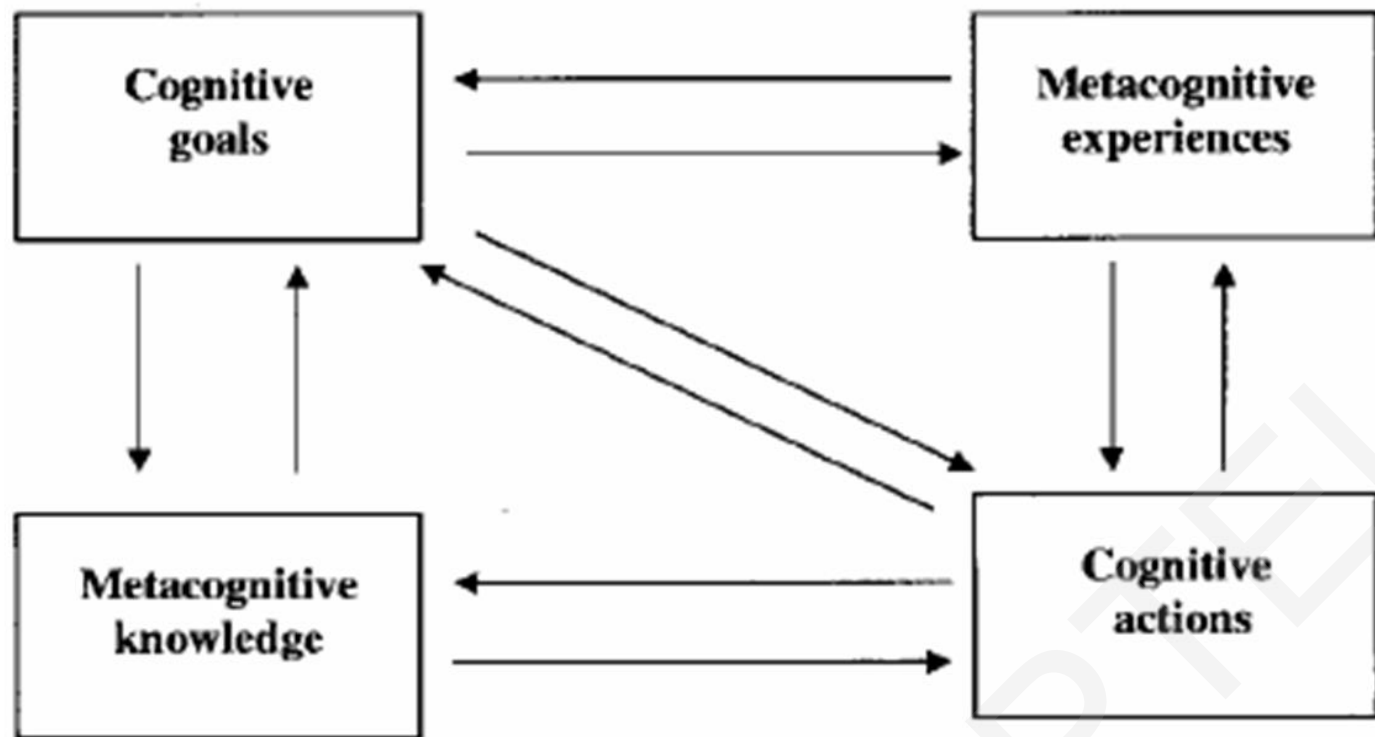


Figure 1. A model of cognitive monitoring. (Source: Flavell, 1981, p. 40.)

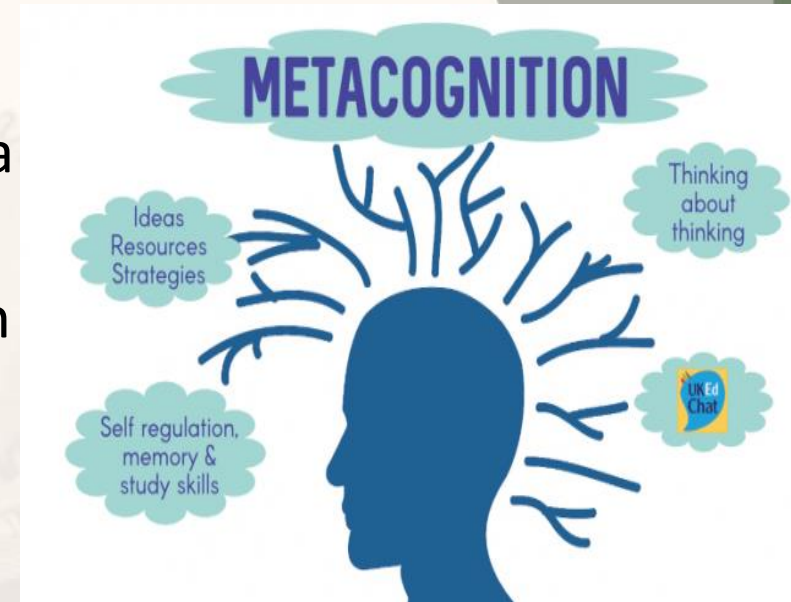
- a) metacognitive knowledge;
- b) metacognitive experiences;
- c) goals (or tasks);
- d) actions (or strategies).

## ➤ Cognitive Task / Strategy

- Ex: Knowledge of finding the sum of a set of numbers
- \* **Cognitive strategies** are basically knowing **how to reach a**
- **goal**, such as how to add the numbers to find the sum (like a study method)
- Ex: remembering things learned earlier that might help with a current task or problem

## ➤ Meta-cognitive Task / Strategy

- Ex: Add the numbers up again
- \* **Meta-cognitive strategies** are to **make sure that the goal** was **reached successfully**, such as **double or triple checking** the correct answer (like a confidence builder)
- Ex: monitoring and directing the processes of problem solving
- **Cognitive Strategy Instruction** enhances Meta-cognition





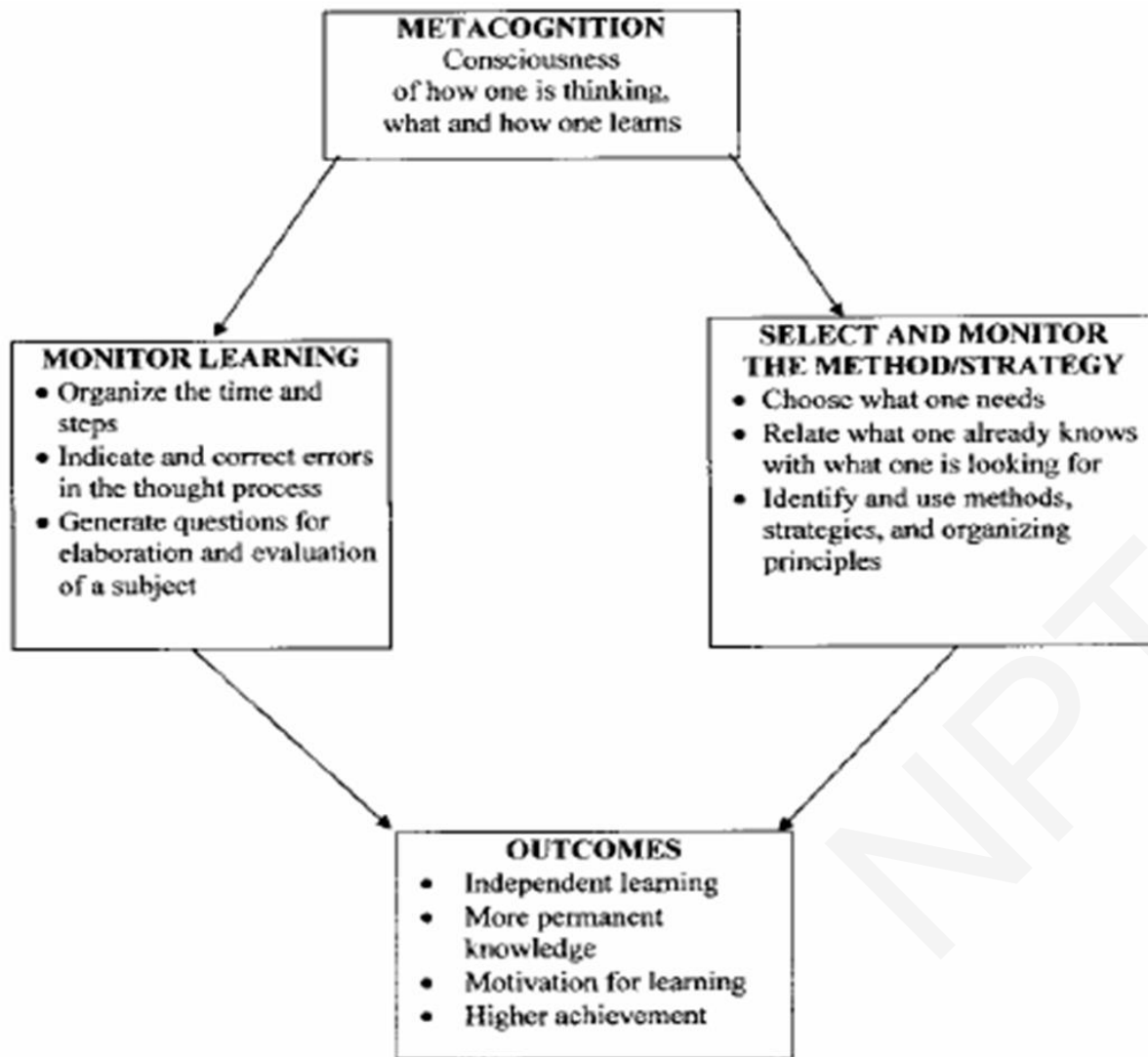
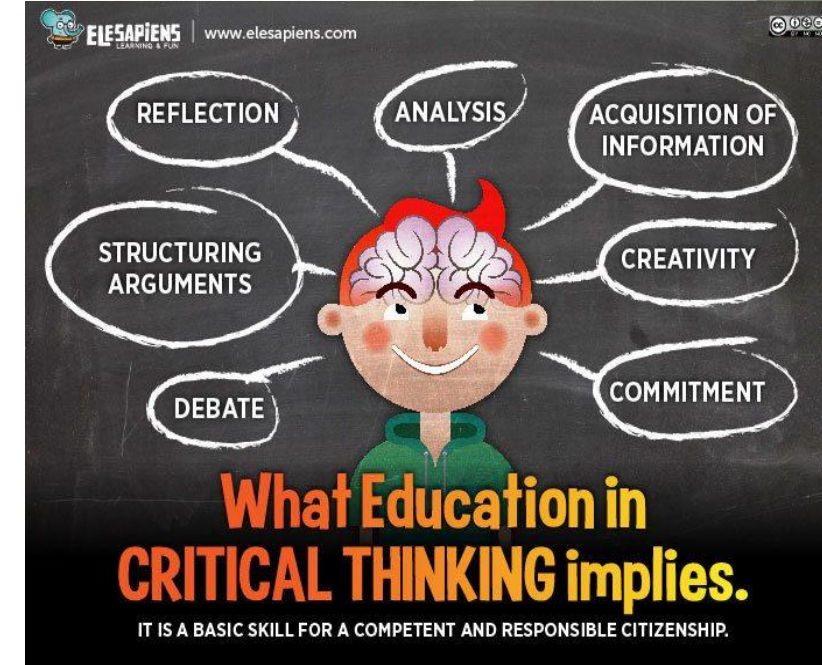
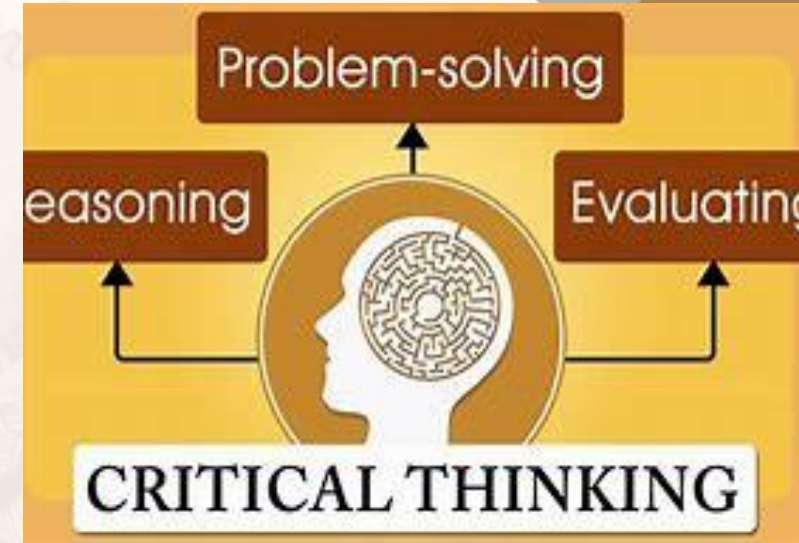


Figure 2. Components of the concept of metacognition. (Source: Koutselini, 1995, p. 51.)



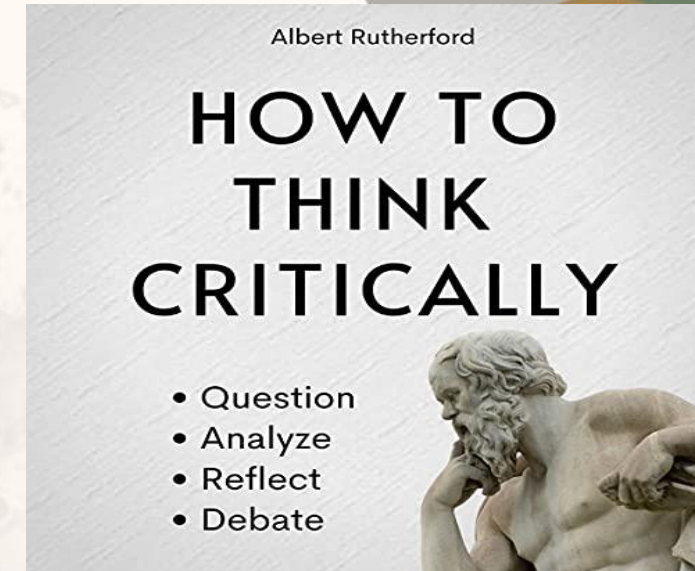
- **Critical Thinking-**

- Critical thinking is the **ability to collect and analyze information to come to a conclusion.**
- It's our ability to parse through **information, data, statistics,** and other details in order to identify a satisfactory solution.
- Developing our critical thinking skills improves our **problem solving skills, boosts our data-driven decision making ability,** and gives us a methodology to tackle complex problems.
- Developing critical thinking is more about learning to **embrace open-mindedness** and bringing **analytical thinking** to our problem framing process.
- **The most important critical thinking skills are:**
- **Analytical thinking:** Part of critical thinking is evaluating data from multiple sources in order to come to the best conclusions.
- **Analytical thinking** allows people to reject bias and strive to gather and consume information **to come to the best conclusion.**





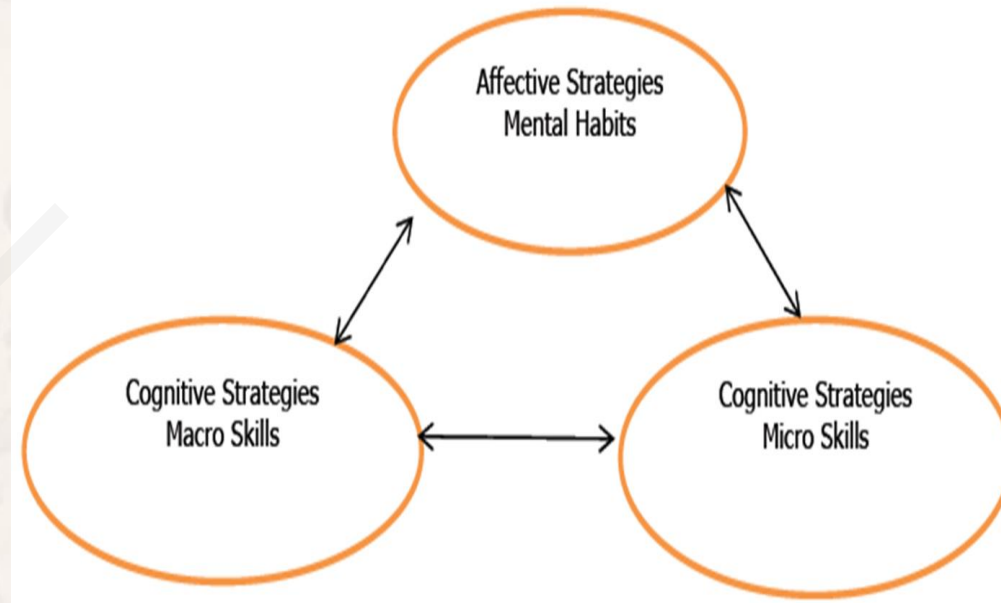
- **Open-mindedness**: This critical thinking skill helps you analyze and process information to come to an unbiased conclusion.
- Part of the critical thinking process is letting your personal biases go and coming to a conclusion based on all of the information.
- **Problem solving**: Because critical thinking emphasizes coming to the best conclusion based on all of the available information, it's a key part of problem solving.
- When used correctly, critical thinking helps you solve any problem—from a workplace challenge to difficulties in everyday life.
- **Self-regulation**: Self-regulation refers to the ability to regulate your thoughts and set aside any personal biases to come to the best conclusion.
- In order to be an effective critical thinker, you need to question the information you have and the decisions you favor—only then can you come to the best conclusion.
- **Observation**: Observation skills help critical thinkers look for things beyond face value.
- To be a critical thinker you need to embrace multiple points of view, and you can use observation skills to identify potential problems.





- **Interpretation:** Not all data is made equal—and critical thinkers know this.
- In addition to gathering information, it's important to evaluate which information is important and relevant to your situation.
- That way, we can draw the best conclusions from the data you've collected.
- **Evaluation:** When you attempt to answer a hard question, there is rarely an obvious answer.
- Even though critical thinking emphasizes putting our biases aside, we need to be able to confidently make a decision based on the data you have available.
- **Communication:** Once a decision has been made, you also need to share this decision with other stakeholders.
- Effective workplace communication includes presenting evidence and supporting your conclusion—especially if there are a variety of different possible solutions.

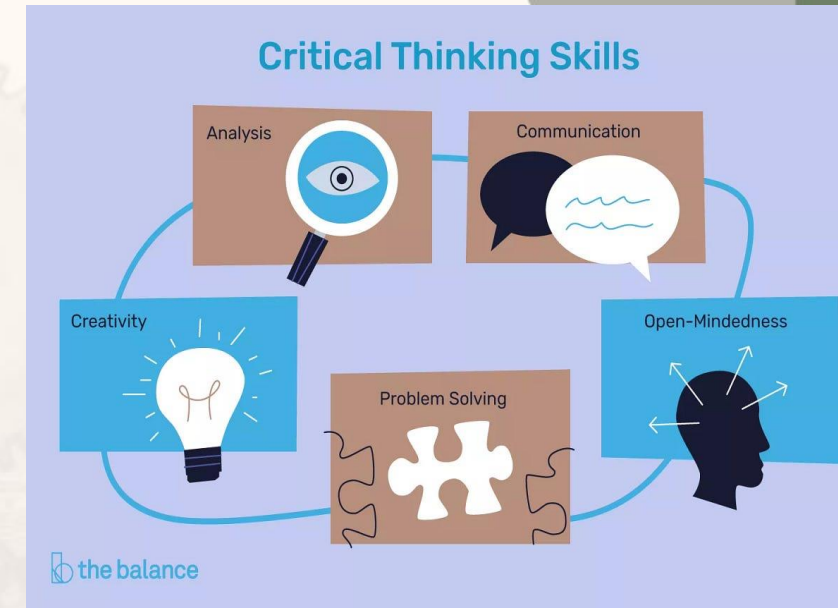
#### Strategies of Critical Thinking



## ❑ STRATEGIES OF CRITICAL THINKING

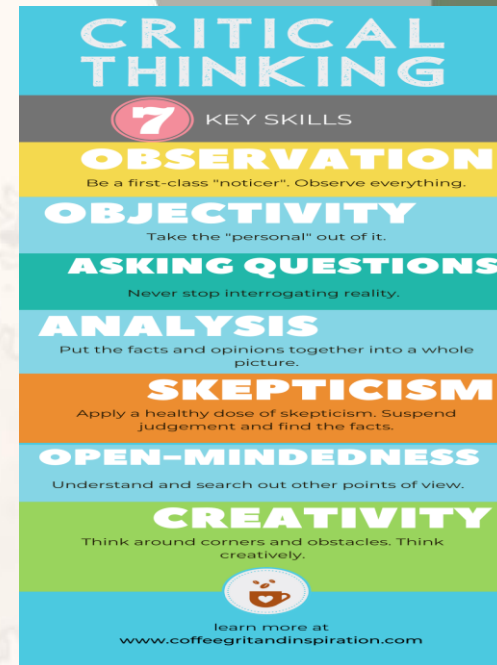
### ➤ Affective Strategies – Mental Habits

- S-1: Independent thinking
- S-2: Developing egocentric or socio-centric insights
- S-3: Realizing objective thinking
- S-4: Understanding the relationship between feelings and thoughts
- S-5: Developing mental modesty and delaying judgement
- S-6: Developing mental bravery
- S-7: Developing mental good faith and honesty
- S-8: Developing mental tenacity
- S-9: Developing trust in thinking skill



## • Cognitive Strategies – Macro Skills

- S-10: Clearing up generalizations and avoiding disturbing their meaning through simplifying
- S-11: Comparing similar states: Transferring insights into new contexts
- S-12: Developing the individual's perspective: Creating/Exploring beliefs, views, theories
- S-13: Clearing up problems, results or beliefs
- S-14: Clearing up and analysing words or word phrases
- S-15: Developing standards for evaluation: Clearing up values and standards
- S-16: Evaluating the credibility of information sources
- S-17: Asking basic and important questions, ensuring the continuity of questions
- S-18: Analysing or evaluating views, interpretations, beliefs, theories





- S-19: Producing solutions or evaluating solutions
- S-20: Analysing or evaluating activities or policies
- S-21: Critical reading, clearing up or probing texts
- S-22: Critical listening
- S-23: Establishing relationships between disciplines
- S-24: Implementing Socratic discussion
- S-25: Thinking in terms of dialogues
- S-26: Dialectical reasoning

### ➤ Cognitive Strategies – Micro Skills

- S-27: Comparing and differentiating real implementations and ideals
- S-28: Thinking perfect about thinking: Using critical vocabulary
- S-29: Paying attention to important similarities and differences
- S-30: Probing and evaluating premises



- S-31: Differentiating related events from unrelated ones
- S-32: **Creating rational inferences**, estimates and remarks
- S-33: **Evaluating evidences** and claimed events
- S-34: **Noticing contradictions**
- S-35: **Exploring products** and results

## **❑ CRITICAL THINKING SKILLS**

- A. **Reliability of Sources**
- B. **Causal Explanation**
- C. **Prediction**
- D. **Generalization**
- E. **Reasoning by Analogy**
- F. **Conditional Reasoning**



- 1. **Evaluating** prejudice and consistency,
- 2. **Differentiating** primary and secondary sources,
- 3. **Evaluating** inferences and their causes,
- 4. **Differentiating** premises, ideas and claims,
- 5. **Noticing** the deficiencies of the argument and uncertainties of explanations,
- 6. **Evaluating** the sufficiency of definitions and suitability of results

- **How to improve Active Thinking Skills-**

- Writing
- Studying
- Mind mapping
- Discussing
- Debating
- Meditating
- Consistent practice





Chart 4: Critical Thinking Skills

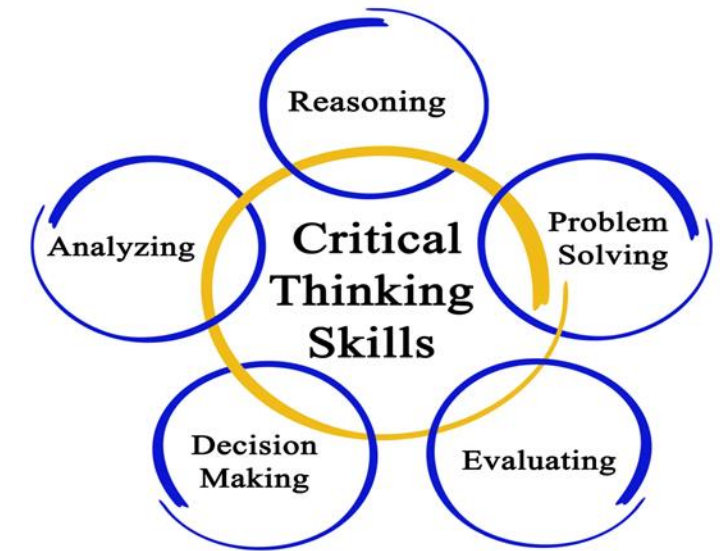
| CRITICAL THINKING SKILLS                                                                                                                                                                    |                                                                                                                            |                                                                                                                                                                                               |                                                                                                                                                          |                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Clearing Up Skills                                                                                                                                                                          | Supporting Skills                                                                                                          | Inference Skills                                                                                                                                                                              | High-level Clearing Skills up                                                                                                                            | Strategy and technical skills                                                                             |
| <ul style="list-style-type: none"> <li>Focusing on a question</li> <li>Analyzing discussions</li> <li>Asking various questions at various levels in order to clear up the states</li> </ul> | <ul style="list-style-type: none"> <li>Judging the reliability of a source</li> <li>Judging observation reports</li> </ul> | <ul style="list-style-type: none"> <li>Being able to make inferences from the current data</li> <li>Being able to think deductively</li> <li>Being able to create value judgements</li> </ul> | <ul style="list-style-type: none"> <li>Being able to define the terms and judge the definitions</li> <li>Being able to determine the premises</li> </ul> | <ul style="list-style-type: none"> <li>Deciding on a movement</li> <li>Interaction with others</li> </ul> |

Source: Prepared with the help of Doğanay, Ahmet and Ünal, Figen (2006:215).



## • Five common and impactful critical thinking skills

- 1. **Observation**-Improving our observation skills by slowing down the pace of processing information and training ourselves to pay closer attention to our surroundings. We may also practice mindfulness techniques, journaling or actively listening in our day-to-day activities.
- 2. **Analysis**-The ability to analyze and effectively evaluate a situation involves knowing what facts, data or information about the problem are important.
- 3. **Inference**-This is a skill that involves drawing conclusions about the information we collect and may require us to possess technical or industry-specific knowledge or experience.
- 4. **Communication-Skills** are important when it comes time to explain and discuss issues and their possible solutions; can be improved through active listening and understanding others' perspectives.
- 5. **Problem-solving-Ability** often requires critical thinking to implement the best solution and understand whether or not the solution is working as it relates to the goal.

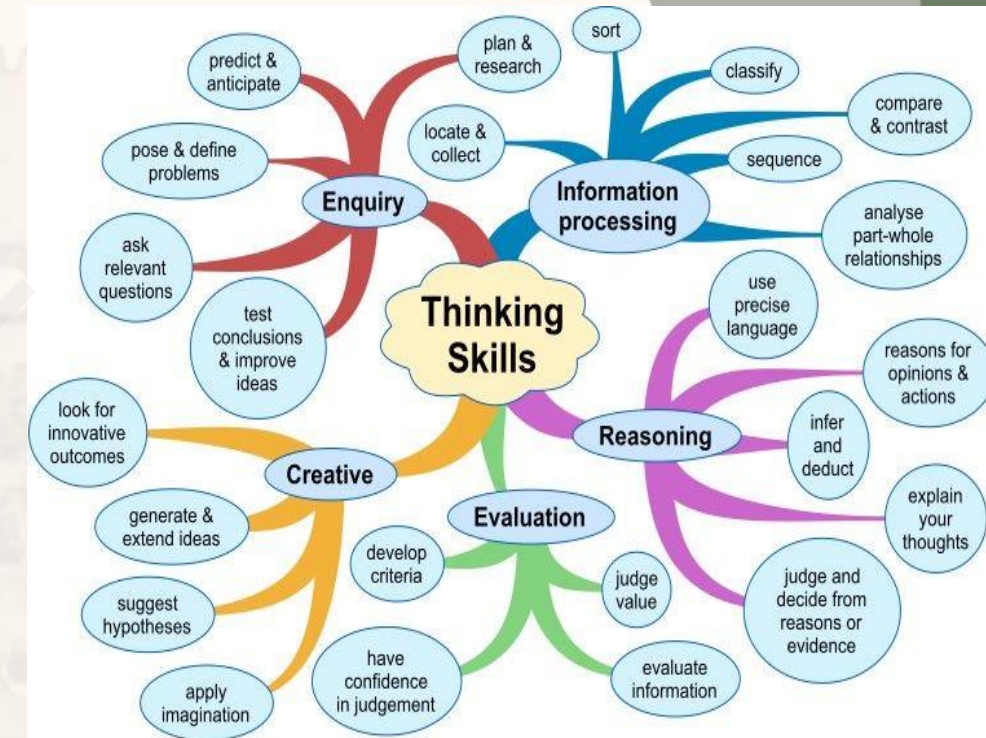




## • How to think critically ???

### • Steps to follow:

- ✓ 1. Identify a problem or issue.
- ✓ 2. Create inferences on why the problem exists and how it can be solved.
- ✓ 3. Collect information or data on the issue through research.
- ✓ 4. Organize and sort data and findings.
- ✓ 5. Develop and execute solutions.
- ✓ 6. Analyze which solutions worked or didn't work.
- ✓ 7. Identify ways to improve the solution.
- ✓ 8. Being objective without allowing personal bias, emotions or assumptions to influence rational thinking.
- ✓ 9. To only analyze a problem based on the context and facts collected after conducting thorough and impartial research.







## Lecture : 19

# Thinking & Cognition

Prof. Atasi Mohanty  
Indian Institute of Technology  
Kharagpur



# Critical & Reflective Thinking

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Critical Thinking</b><br/> <b>Purpose:</b> Critical Judgement<br/> <b>Basic Skill:</b> Evaluating the ratioanlity of thoughts</p> <ol style="list-style-type: none"> <li>Evaluating the basic information <ul style="list-style-type: none"> <li>Accuracy of observation</li> <li>Credibility of sources</li> </ul> </li> <li>Inference <ol style="list-style-type: none"> <li>Usage of evidences <ul style="list-style-type: none"> <li>Causal explanation</li> <li>Guess</li> <li>Generalization</li> <li>Explanation through simulation</li> </ul> </li> <li>Deduction <ul style="list-style-type: none"> <li>Conditional reasoning (If...then)</li> </ul> </li> </ol> </li> </ol> <p><b>Model attitude:</b> We should take healthy reasoning as a basis and be open-minded.</p> | <p><b>Creative Thinking</b><br/> <b>Purpose:</b> Original production<br/> <b>Basic Skill:</b> Skill of producing an idea</p> <ol style="list-style-type: none"> <li>Producing possibilities <ul style="list-style-type: none"> <li>Variety of ideas (fluency)</li> <li>Changeability of ideas (flexibility)</li> <li>New ideas (originality)</li> <li>Detailedness of ideas (details)</li> </ul> </li> <li>Creating metaphors <ul style="list-style-type: none"> <li>Analogy/metaphor</li> </ul> </li> </ol> <p><b>Model attitude:</b> Considering unusual ideas</p> | <p><b>Clearing up and Understanding Opinions</b><br/> <b>Purpose:</b> Understanding thoroughly and remebering correctly<br/> <b>Basic Skill:</b> Clearing up ideas</p> <ol style="list-style-type: none"> <li>Analyzing ideas <ul style="list-style-type: none"> <li>Comparing similarities and differences</li> <li>Classifying / describing</li> <li>Finding meronymy</li> <li>Determining sequency</li> </ul> </li> <li>Analyzing evidences <ul style="list-style-type: none"> <li>Finding cause and effect relation</li> <li>Finding unspecified assumptions</li> </ul> </li> </ol> <p><b>Model attitude:</b> Looking for clarity and related information</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



Chart 2.1: The relationship between thinking skills

| Types of thinking | Purposes                                           | Thinkings Skills                                                             | Relations between them                                                                                                                            |
|-------------------|----------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Critical Thinking | Evaluating the opposite states or clarity of ideas | Defining states or ideas, analyzing opposite views, evaluating evidences     | Critical thinking is needed for all thinking skills.                                                                                              |
| Creative Thinking | Producing new ideas and products                   | Determining ideas, re-structuring the problem, determining the possibilities | The newly-created product which has been produced through creative thinking is evaluated by means of critical thinking.                           |
| Making a Decision | Making an informed decision                        | Thinking the information, defining the alternatives and making a decision    | Critical thinking is one of the basic processes necessary for making a decision.                                                                  |
| Problem Solving   | Finding one or more solutions for a problem        | Defining, explaining, choosing, implementing and evaluating a strategy,      | While problem solving starts with a problem, critical thinking encompasses evaluation of all information, ideas and events that one comes across. |

Definition



Experience something.

Think about what happened.

Learn from the experience; that's reflective thinking.

Source: Prepared with the help of Mehmet Kaan Demir, (2006:43).



## ☐ Reflective Thinking

- Critical thinking is a broader concept, which includes reflective thinking
- Reflective thinking has a very close relationship with meta-cognitive thinking
- In reflective thinking, the individual thinks about his/her thinking and learning styles
- Reflective thinking is a skill which can be displayed in parallel with the development of individual's self-regulation processes
- Organizing, reasoning, developing assumptions and predictive skills of critical thinking skills are within the scope of questioning and evaluation skills of reflective thinking



## ➤ DEVELOPING REFLECTIVE THINKING

- 1. Writing Learning Texts
- 2. Keeping Reflective Diary
- 3. Preparing Portfolio
- 4. Constructing Concept Maps
- 5. Asking Question
- 6. Self-questioning
- 7. Joining Reflective Discussions
- 8. Negotiated Learning
- 9. Self-evaluation
- <https://www.youtube.com/watch?v=vg936IW9i7Q>



- **What is Creative thinking?**
- Creative thinking is the **process of nurturing our imagination** allowing us to “**think out of the box**”.
- Creative thinking is the ability to consider something in a **new way**.
- **Creative thinking can involve:**
  - A **new approach** to a problem
  - A **resolution to a conflict** between employees
  - A **new result** from a data set
  - A previously untried approach to earn revenue
  - A **new product—or product feature**
- Creative thinking isn't limited to artistic types; it is **a skill that anyone can nurture and develop**.







- Creative thinking requires a compilation of analytical, problem-solving, organizational and communication skills to evaluate and solve a problem.
- Creative thinking is all about developing innovative solutions to problems.
- Creative thinkers brainstorm not only a large number of ideas but also a variety and range of them.
- How to think creatively: Steps
- Gather what information you already have.
- Consider the obvious solution or process.
- Brainstorm additional solutions.
- Consider how the topics connect.
- Apply the solutions





- “Everyone is born creative” – Hugh Macleod
- Everyone Should Become a Creative Thinker
- Thinking creatively adds value not only to our life but also to the life of others.
- It helps us become self relying and more confident,
- Thinking open-mindedly gives us a reason to wake up every morning to start exploring new ideas and thoughts; it makes our life worth living.
- It can bring out hidden talents that we might have not noticed before.
- Thinking ‘out of the box’ makes us feel satisfied at all phases of our life.
- The more experienced we get with thinking creatively, the easier it becomes to master difficult situations we may face in our life.

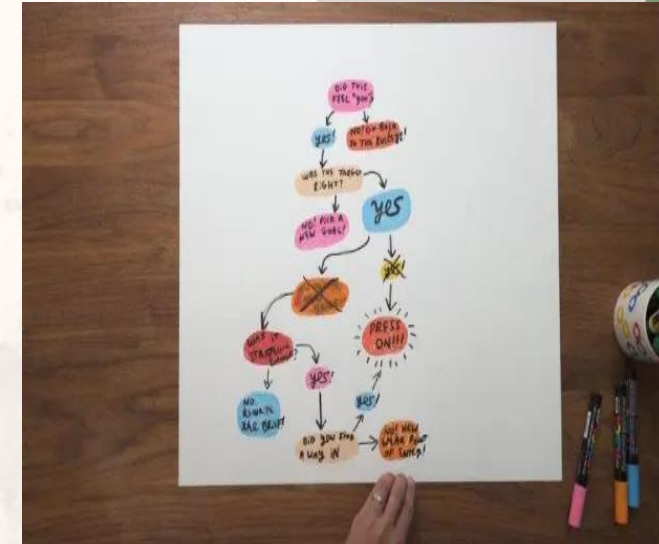
- **How to apply creative thinking in Learning Process-**
- Risk Taking behavior
- Playing mental games & puzzles
- Regularly asking why, how, for understanding new information,
- Generating large quantity of ideas by thinking divergently for producing a big number of possible solutions to a critical problem.
- Day dreaming, imagination
- Reading a lot of stuff & travelling to new places, knowing different cultural practices/languages.
- Adopting Lateral thinking-It is expanding our thinking limits, freeing ourselves from boundaries, giving us the chance to explore things we never knew existed;
- Allowing our mind to discover new options; Looking beyond normal alternatives ,searching for diverse strategies, resources.





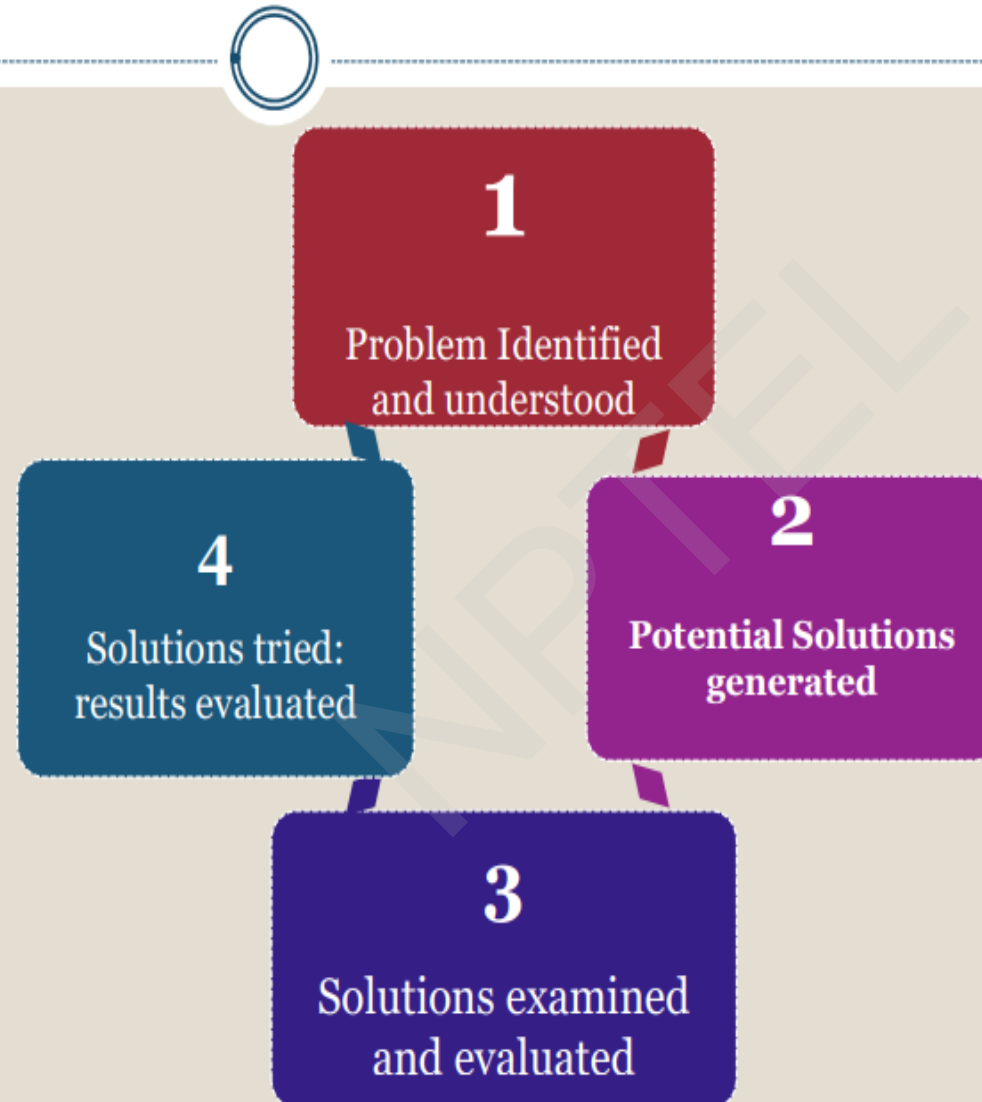
- **Stages of the Creative Thinking Process-**

- 1. Preparation: The Inspiration Phase
- 2. Incubation: Absorbing and Processing
- 3. Insight: The “Eureka” Moment
- 4. Evaluation: Putting our Idea Through the Wringer
- 5. Elaboration: Putting Pen to Paper for details



# Problem Solving

- Efforts to develop or choose among various responses in order to attain desired goals.



## Methods of solving problems

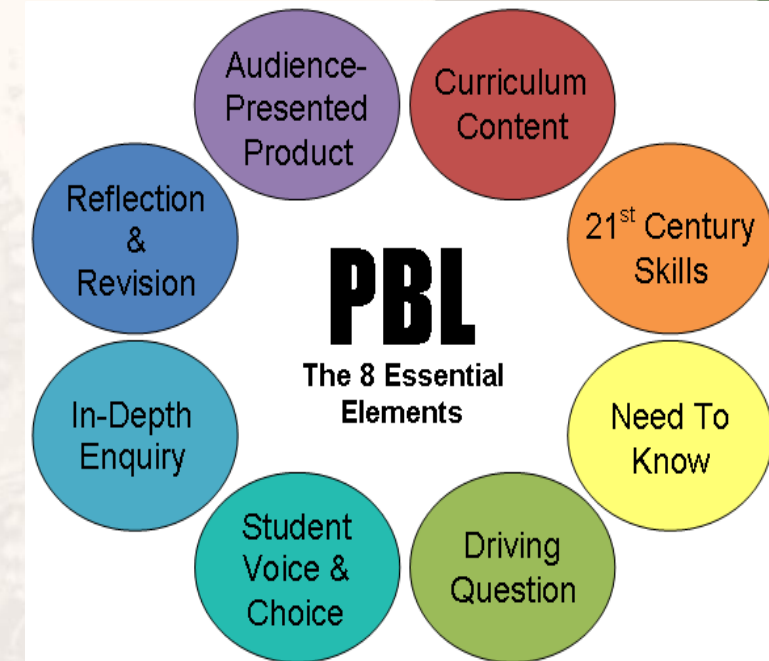


- **Trial and error:** A method of solving problems in which possible solutions are tried until one succeeds
- **Algorithm:** A rule that guarantees a solution to a specific type of problem.
- **Analogy:** A strategy for solving problems based on applying solutions that were previously successful with other problems similar in underlying structure.



## ❑ Problem Solving & PROBLEM-BASED LEARNING (PBL)

- PBL is a form of **constructivist learning**
- **Exposes** students to a **complicated situation**
- PBL represents a **learning based on experience** aiming to up skill students in **learning how to learn** and to **increase their capacity for learning**
- Students **define the real problem** and learn whatever is necessary to work out a **valid solution by means of research**
- It **develops skills** and motivation and maintains the **skill for lifelong learning**



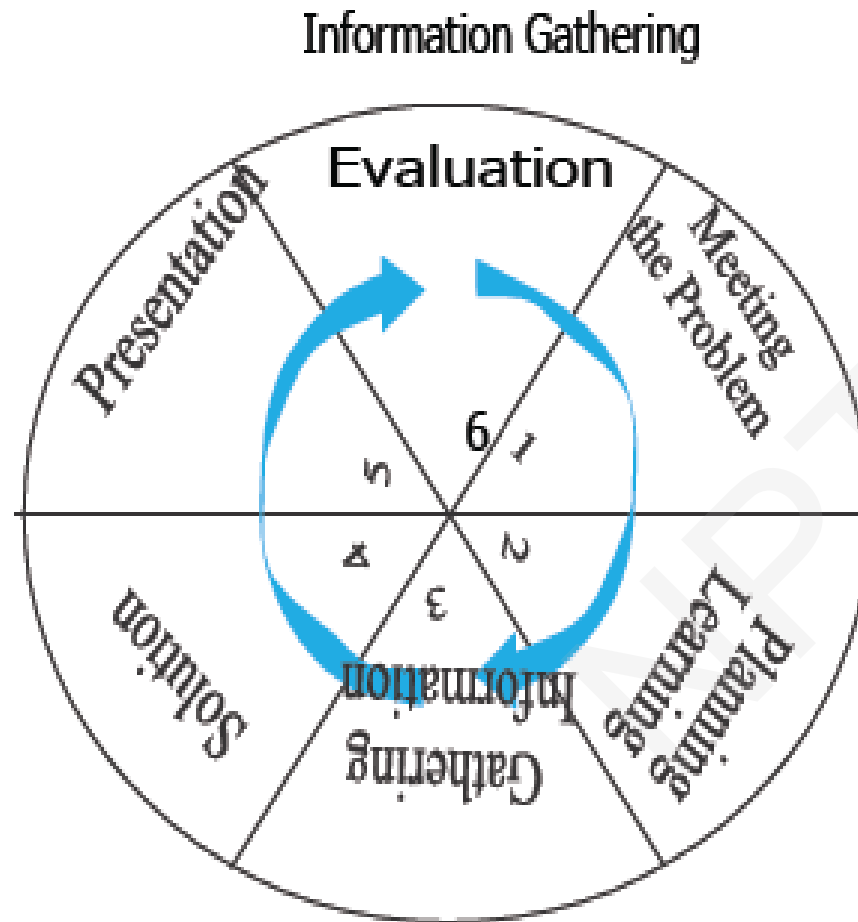
# CHARACTERISTICS OF PROBLEM-BASED LEARNING

## ✓ The Steps for the Practice of Problem-based Learning

- Meeting the problem and defining it
- **Determining** what one knows and what he needs to know and **putting his thoughts into order**
- **Gathering** and **sharing information**
- **Producing potential solutions** and **hypotheses**
- Determining the **learning issues**
- The **application of the new knowledge** to the problem and **re-evaluation**
- Identifying the best solution
- **Explaining** the problem and its solution briefly
- **Evaluation** and **presentation** of new learning



The practice process of problem-based learning has been given in six steps



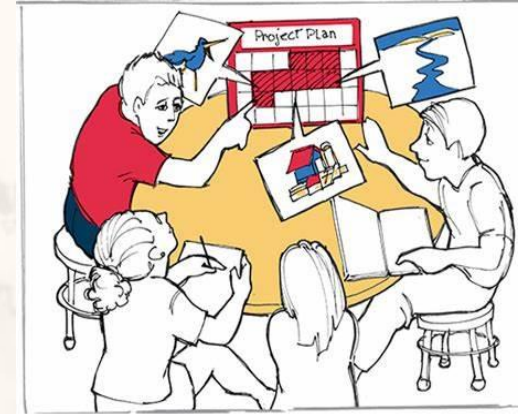
**Diagram 1:** Practice process of PBL



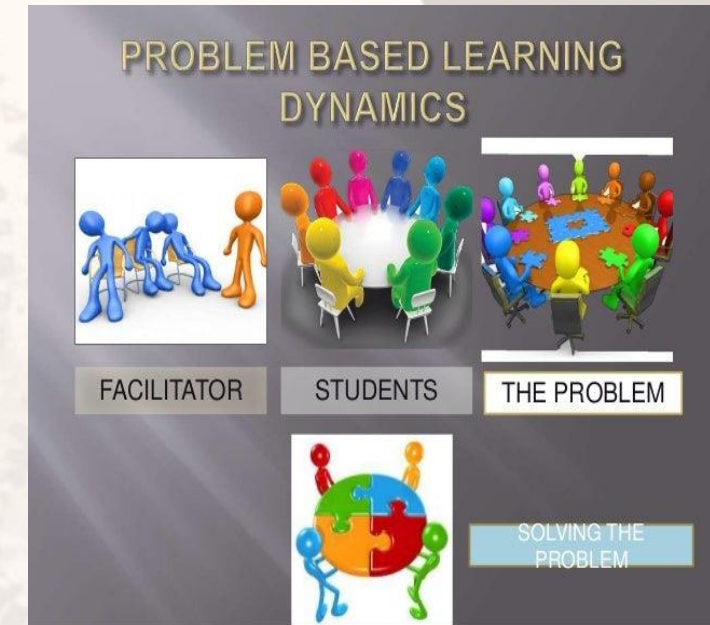


## ➤ The Role of the Learner in PBL

- Analyzing the problem situation by comprehending the structure of the problem broadly
- Developing practical solutions to the problem
- Undertaking a decision-making role in debates within the group
- Cooperating with the tutor and friends and determining the learning goals, which need researching for the solution to the problem
- Identifying the resources and strategies, which will be able to convey to the learning goals
- Assessing the conclusions one draws from the data obtained, that is, the learning products
- Checking both the learning goals and the learning media
- Presenting his opinions in a clear and understandable way in which the other group members can conceive

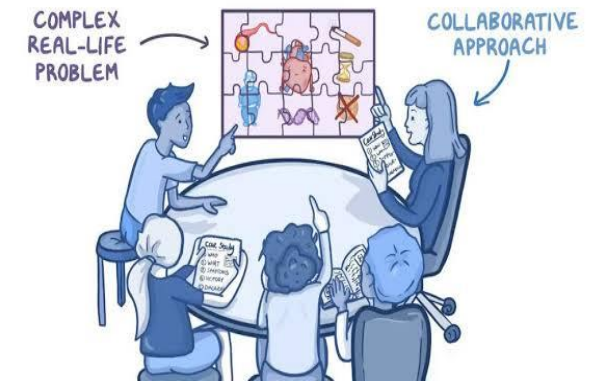


- Evaluating different views impartially
- Being aware of his roles and responsibilities individually
- Exhibiting behaviour of a type, which defends new ideas and situations and gets them across to the other members
- Congratulating and appreciating the other group members on account of their convenient suggestions leading to the solution
- Developing the knowledge infrastructure for the solution to the problem
- Being able to set certain targets, which may be useful in using suitable research procedures and directing the group
- Carrying out observations and practices taking into consideration the processes used by the implementers of the field or the discipline related to the problem



- **Becoming the instructor** of each other by teaching what they have learned to the other group members
- **Metacognitive processing**: An expanded level of awareness that allows us, in a sense, to observe ourselves in the problem solving process
- Being **courageous in focusing** on a new problem and on its potential solution
- Getting into **contact with the external world** and other people so as to discover concepts and to use his abilities
- **Approaching problems** and their solutions with new and original ideas
- **Being inquisitive** during the process

## PROBLEM-BASED LEARNING (PBL)





- Giving positive input to constructive criticisms
- Joining group work on time and on a regular basis
- Assessing his and his friends' contribution to the group work in the course of it
- Finding out important resources for the solution to the problem and sharing them
- Turning the work into a report and presenting it to the class





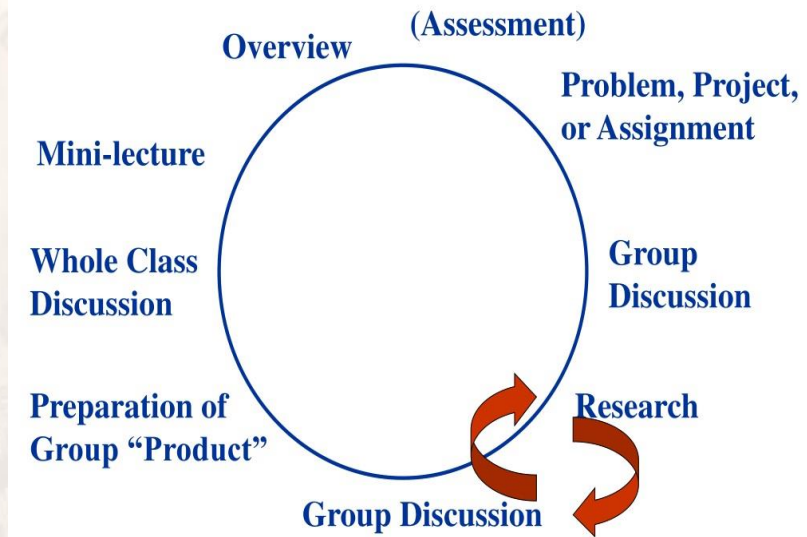
Table 2: Differences between traditional approach and problem-based learning

|                       | <b>Traditional Approach</b>                                                                                                                                                                 | <b>PBL Approach</b>                                                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>      | Enabling students to repeat the knowledge they have learned as it is when they are asked to.                                                                                                | Enabling students to construct their knowledge themselves to work out a solution to a problem situation.                                                                                                                                                                                  |
| <b>Tutor's Role</b>   | 1.The tutor gives information and directs their thinking.<br>2.The tutor guides students into learning and evaluates them.<br>3.Learning materials are prepared and presented by the tutor. | 1.The tutor, as a cognitive guide, exposes students to a problem<br><br>2.The tutor, as a resource person, asks students questions, relates to students' world and directs<br><br>3.The tutor identifies the learning situations and students select the problems and learning materials. |
| <b>Student's Role</b> | 1.The student is passive.<br><br>2.As a follower, the student awaits the leadership and guidance of the tutor.                                                                              | 1.As problem solvers, students work out various solutions to problems they meet making use of the resources available.<br>2.As participants, students are active in the learning process and research the problem extensively.                                                            |
| <b>Information</b>    | It is gathered, organized and presented to students by tutor.                                                                                                                               | Very little of the information is presented by the tutor. Most of it is gathered and constructed by students.                                                                                                                                                                             |

## ➤ ADVANTAGES OF PROBLEM-BASED LEARNING

- ✓ The lesson is **student-centred** rather than teacher-centred
- ✓ It develops **self-inspection** of students
- ✓ It **joins practice and theory**
- ✓ It maintains **life-long learning**
- ✓ It provides **active learning**
- ✓ It gains **group-working skills**
- ✓ It gains **problem-solving skills**
- ✓ It **increases science literacy**
- ✓ It gains **scientific process skills**
- ✓ It gains **highly memorable knowledge**
- ✓ It gains **meta-cognitive skills**

### The Problem-Based Learning Cycle







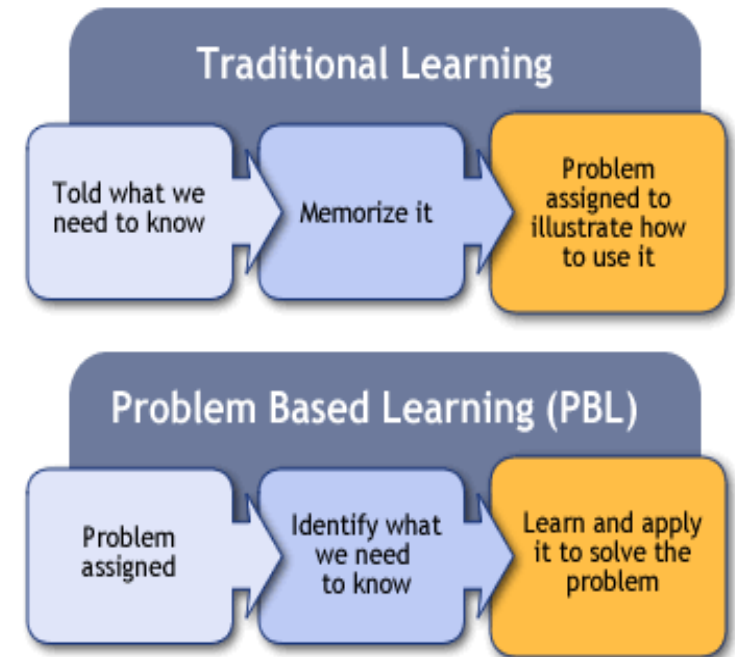
## Lecture : 20

# Thinking & Cognition

Prof. Atasi Mohanty  
Indian Institute of Technology  
Kharagpur



- It gains self-directed learning skills
- It gains critical thinking skills
- It gains cooperative learning skills
- It facilitates high motivation and positive attitude
- It gains communication skills
- It gains higher-order thinking skills
- It constitutes a model for teacher candidates
- It increases skills in using information resources
- It develops decision-making ability



# 21<sup>st</sup> Century Learning Environment



## New Learning Environments for the 21<sup>st</sup> Century



Exploring the edge



ah, imagining  
new possibilities.

Work with, not  
against, change.





# New Learning Models



The architecture studio--  
the atelier form of learning  
a powerful social learning environment



where work  
in progress  
is made public.

learning as  
enculturation  
into a practice.

Does this generalize or scale?

# Technology Enabled Active Learning



MIT's studio for 8.02 electricity and magnetism --  
technology enabled active learning (teal)  
following RPI's studio physics



moving seamlessly between lecture, experiment & discussion  
and addressing the drop out rate

# Learning to be



'Learning to be' sooner rather than later!







## Learning to be – in an epistemic frame

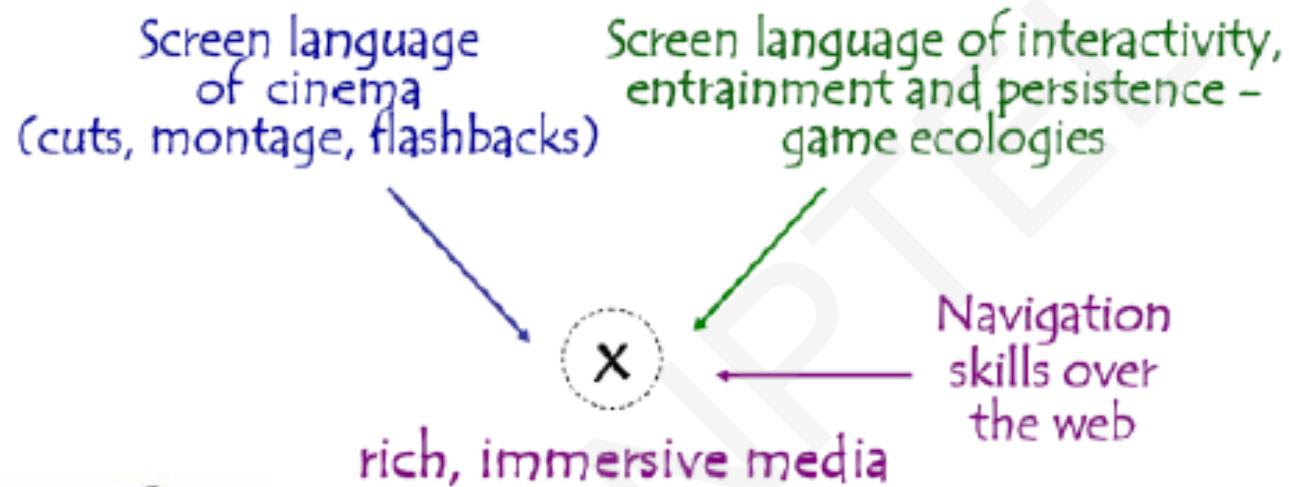
Enculturating into the practices of a field often via  
legitimate peripheral participation – apprenticeship  
a way of seeing  
a way of knowing  
sensing what constitutes an interesting problem  
knowing what constitutes an elegant solution  
being able to engage in productive inquiry

*Productive inquiry is that aspect of any activity  
where we are deliberately (though not always  
consciously) seeking what we need, in order to do  
what we want to do – e.g. leveraging the net.*

(productive inquiry also as active leisure)



## The Emerging Vernacular of the Digital Age and the challenge of fostering multi-media literacy



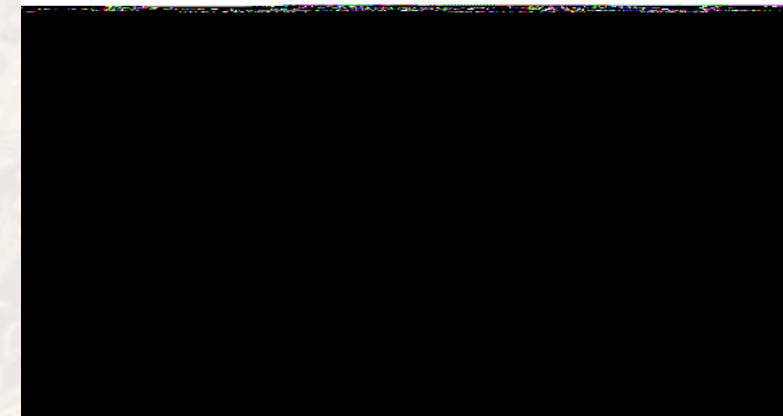
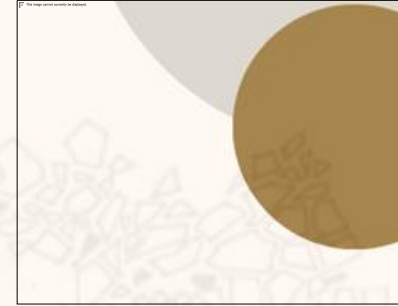
Remember, exposure doesn't mean you can really  
both read and write – critical skills for 21<sup>st</sup> Century:  
democracy & journalism.

# Concept Mapping

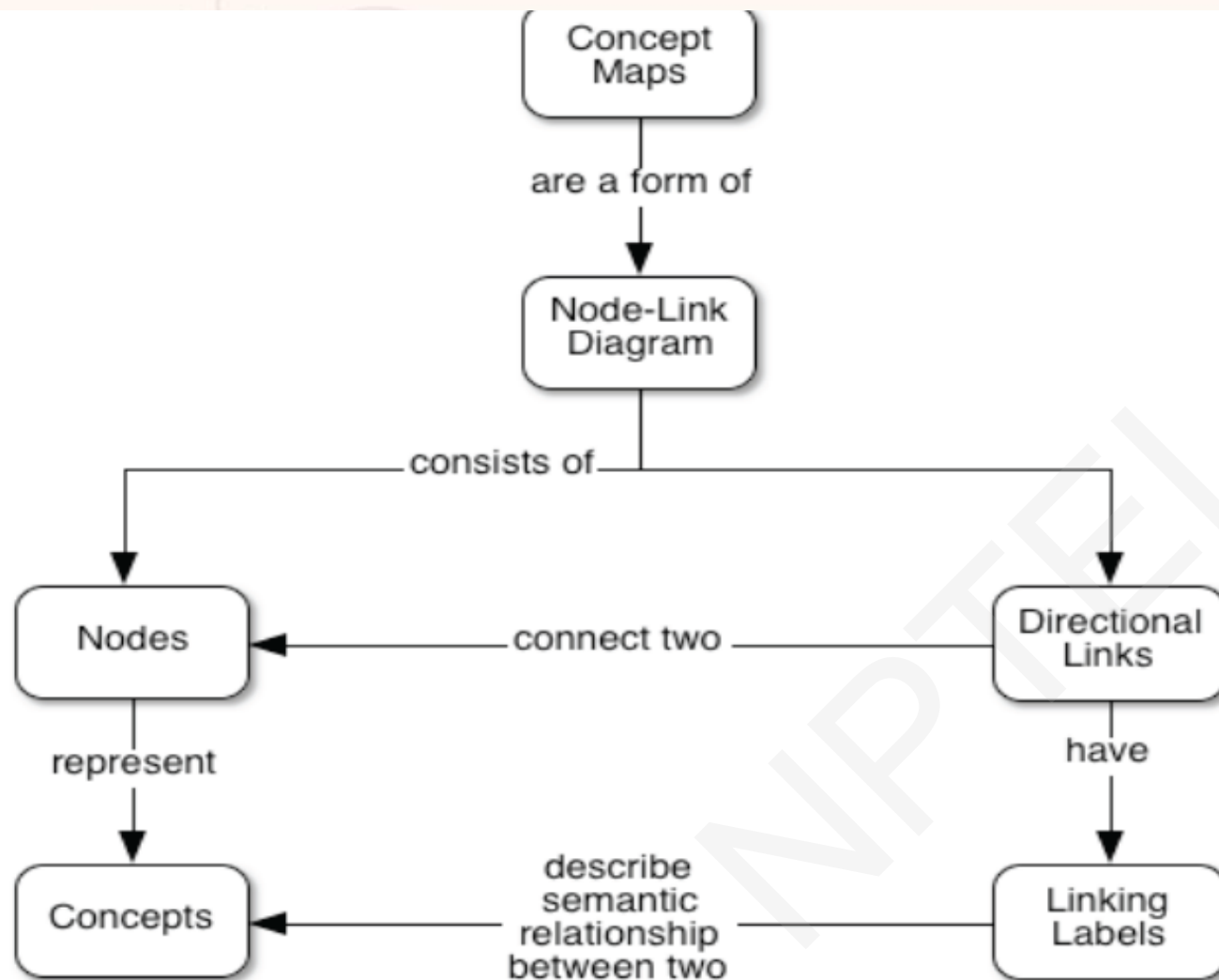


## ❏ DEFINITION:

- A concept map is a **node-link diagram** showing the semantic relationships among concepts
- The technique for constructing concept maps is called "**concept mapping**"
- A concept map consists of **nodes, arrows as linking lines, and linking phrases** that describe the relationship between nodes
- **Two nodes connected** with a labelled arrow are called a **proposition**
- Concept maps are **versatile graphic organizers** that can **represent many different forms of relationships** between concepts

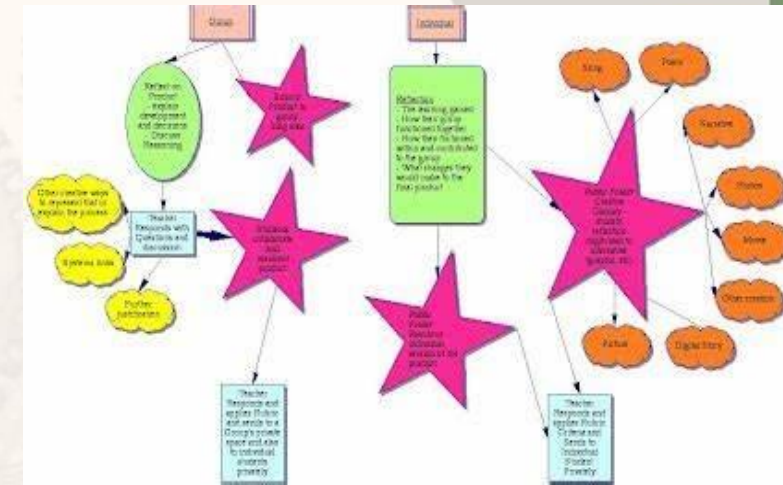






BACKGROUND: The theoretical framework of concept mapping is based on David Ausubel's assimilation theory of learning [See Ausubelian Theory; Meaningful Learning],

- A concept map is a type of **graphic organizer** used to help students **organize** and **represent knowledge** of a subject
- Concept maps **begin with a main idea (or concept)** and then **branch out** to show how that **main idea can be broken down into specific topics**
- Concept mapping is a powerful way for students to **reach high levels of cognitive performance**
- A concept map is also not just a learning tool, but an **ideal evaluation tool** for educators measuring the growth of and **assessing student learning**
- As students create concept maps, they **reiterate ideas** using their own words and help **identify incorrect ideas** and concepts



## ➤ Benefits of Concept Mapping

- ✓ Concept mapping serves several purposes for learners:
  - Helping students **brainstorm** and **generate new** ideas
  - Encouraging students to **discover new concepts** and the propositions that **connect them**
  - Allowing students to more clearly **communicate ideas, thoughts** and **information**
  - Helping students **integrate new concepts** with older concepts
  - Enabling students **to gain enhanced knowledge** of any topic and evaluate the information







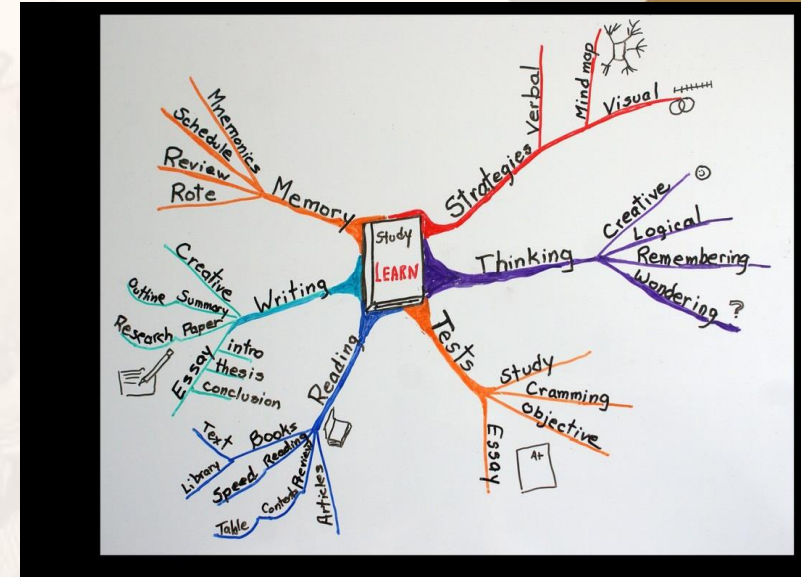
## ➤ How to Build a Concept Map

### ✓ Start with a main idea, topic, or issue to focus on

- A helpful way to determine the context of your concept map is to choose a **focus question**—something that needs to be solved or a conclusion that needs to be reached
- Once a topic or question is decided on, that will help with the **hierarchical structure** of the concept map

### ✓ Then determine the key concepts

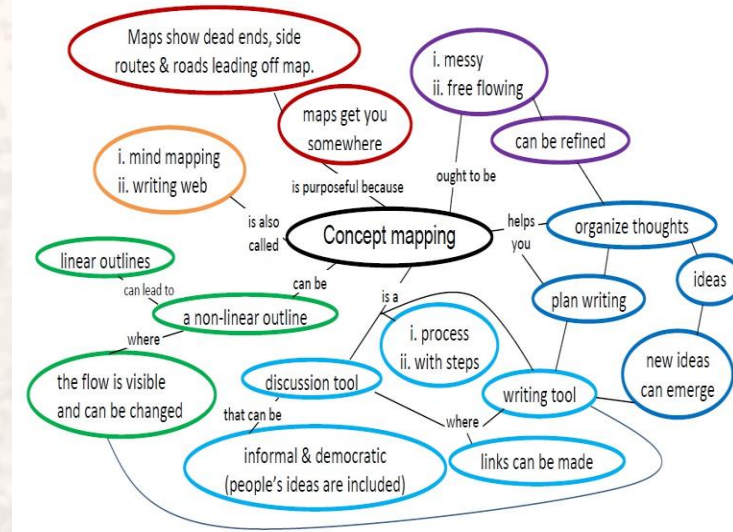
- Find the **key concepts that connect and relate to your main idea and rank them**; most general, inclusive concepts come first, then link to smaller, more specific concepts.





## ✓ Finish by connecting concepts--creating linking phrases and words

- Once the basic links between the concepts are created, add cross-links, which connect concepts in different areas of the map, to further illustrate the relationships and strengthen student's understanding and knowledge on the topic
- A concept map is never finished
- After a preliminary map is constructed, it is always necessary to revise this map
- Other concepts can be added by student under the guidance of teacher in classroom work
- Good maps usually result from several revisions



## Kinds of Concept Mapping

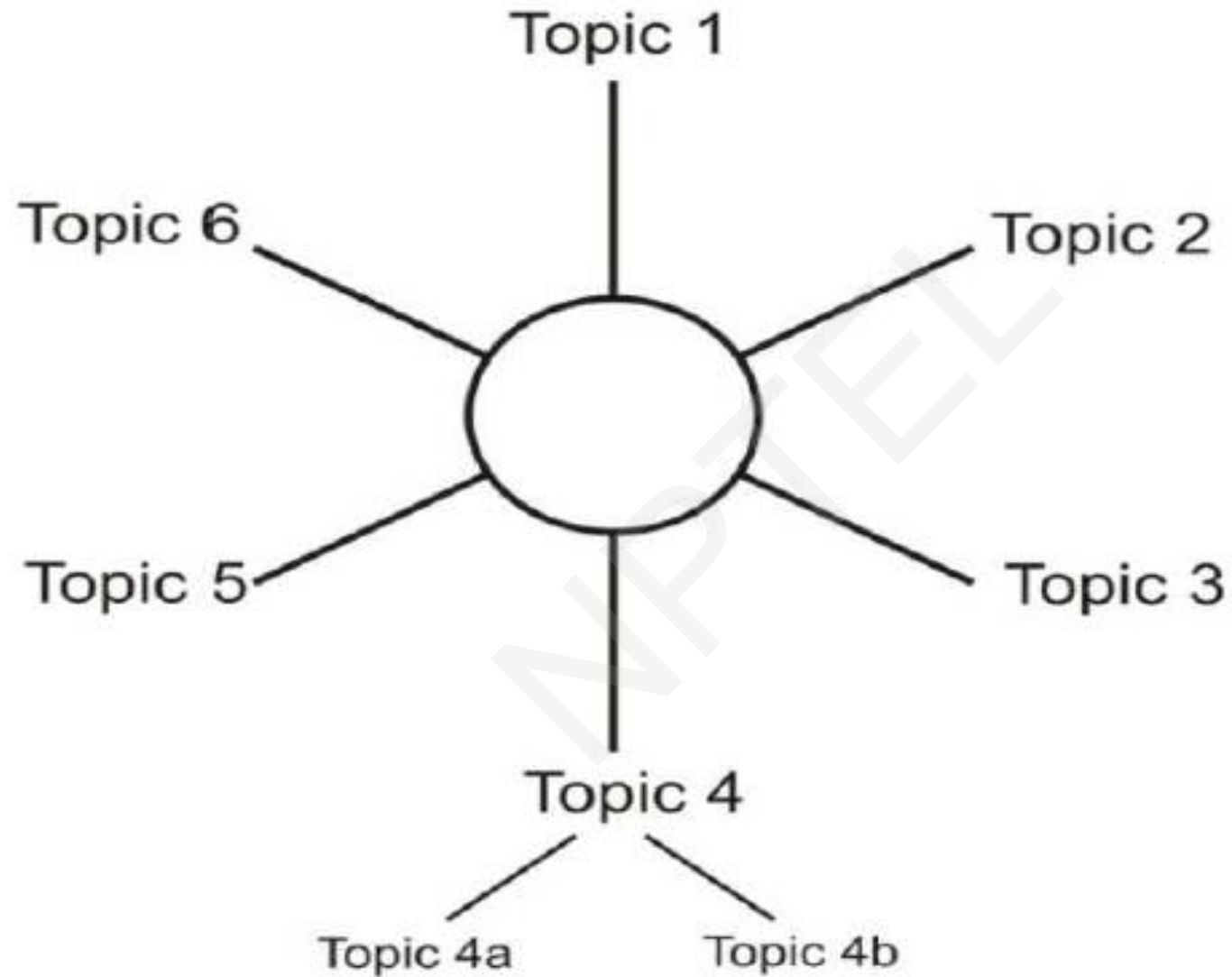


Figure 2a: Spider Concept Map



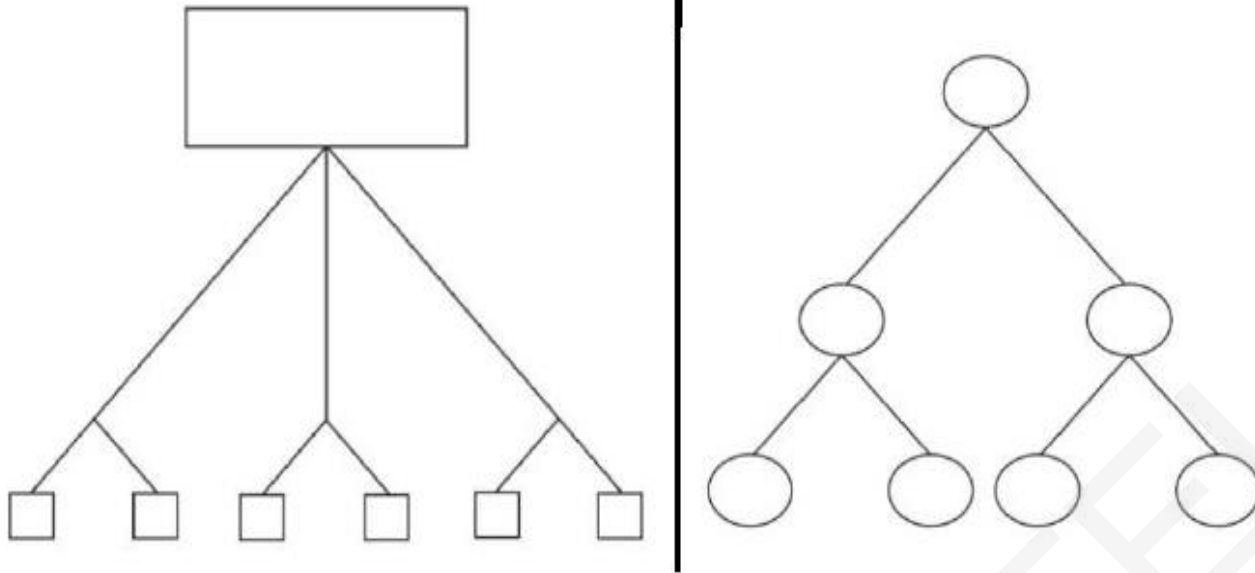


Figure 3a: The hierarchy Concept Map

3. The flowchart concept map organizes information in a linear format.

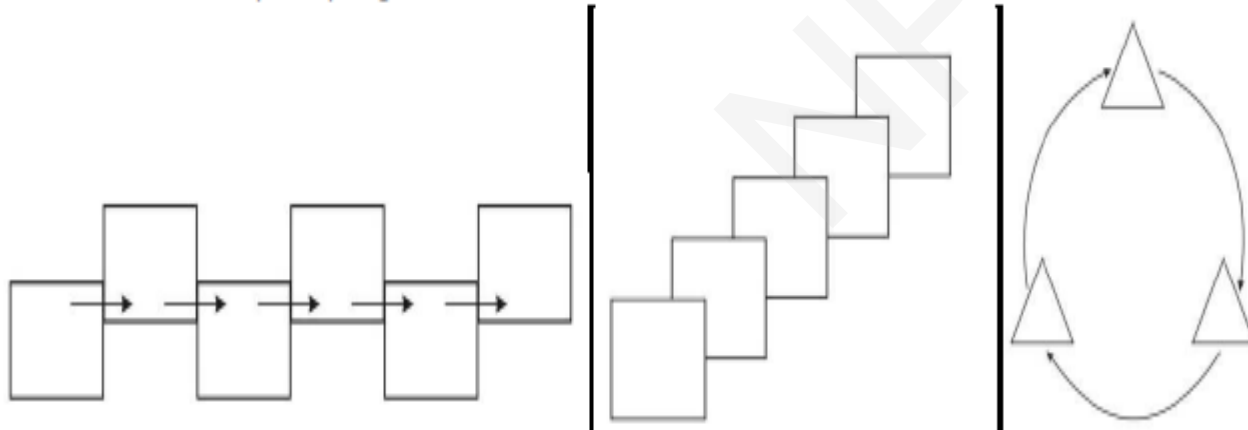


Figure 4a: The flowchart Concept Map

4. The systems concept map organizes information in a format. Includes all data on the map and shows many relationships between the data. Uses critical thinking skills along with problem solving skills.

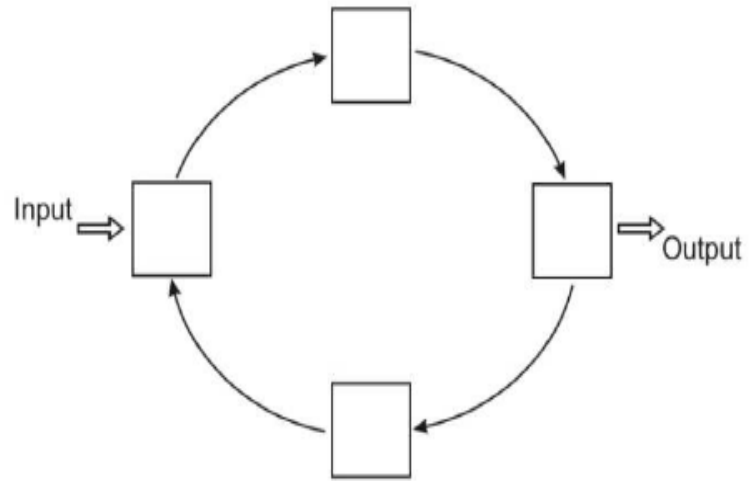


Figure 5a: The systems Concept Map

5. Multi dimensional (3D dimensional) concept map describes the flow or state of information or resources which are too complicated for a simple two-dimensional map.

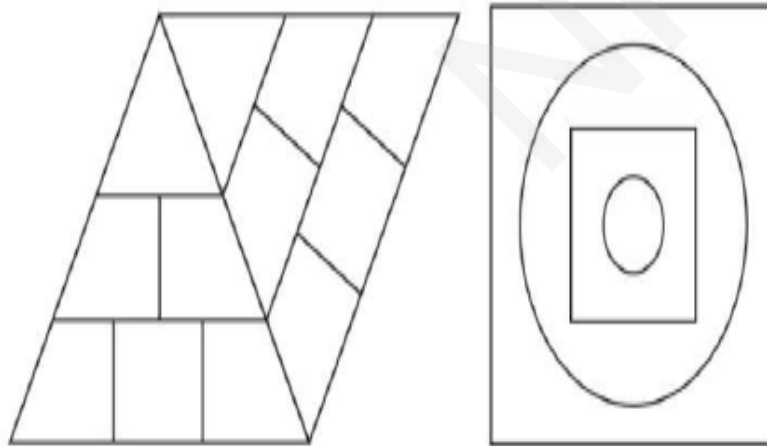


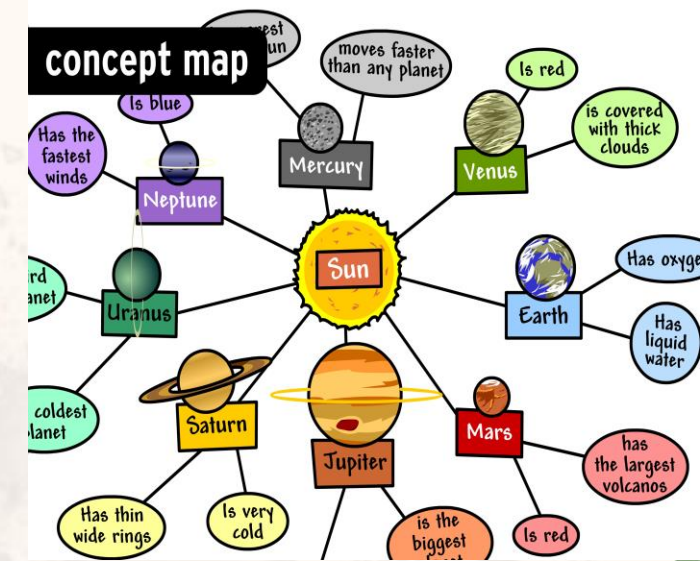
Figure 6a: Multi dimensional (3D dimensional) concept map





- **Concept maps as Advance Organizers**

- “Ausubel suggests that **advance organizers** might foster **meaningful learning** by prompting the student regarding pre-existing super-ordinate concepts that are already in the **student's cognitive structure**, and by otherwise **providing a context of general concepts** into which the **student can incorporate progressively differentiated details**.
- Ausubel claims that by presenting a global representation of the knowledge to be learned, **advance organizers** might foster "**integrative reconciliation**" of the sub-domains of knowledge - the ability to understand **interconnections** among the basic concepts in the domain.”
- (Ausubel's Advance Organizers  
(<http://www.csudh.edu/dearhabermas/advorg03.htm>)





## ➤ Cognitive maps as concept maps

- Läge et al. (2008) suggest a strategy to support assimilative learning-
- Assimilative is understood as integrating new information into existing knowledge or cognitive structures without restructuring the current schema
- A stable and most notably a correct memory representation which "spans" the knowledge space is essential
- Building elaborately a basic structure as well as the assimilative integration of new information can be eased with the aid of cognitive maps

### LESSON SUMMARY

#### cognitive map

*a mental picture or image of the layout of one's physical environment*



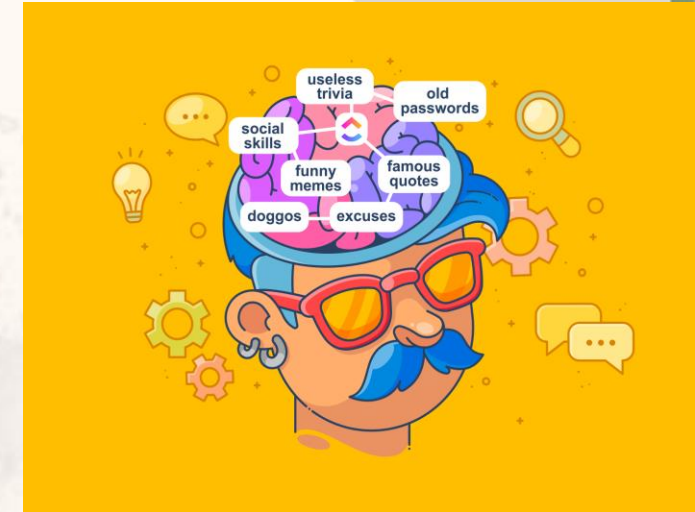
©Study.com

## ➤ Free Mind mapping software

- See also mind map
- **Mindmeister** (<http://www.mindmeister.com>)
- bubbl.us (<http://www.bubbl.us/>)
- Mindomo (<http://www.mindomo.com/>)
- **Tex2Mindmap** (<http://www.text2mindmap.com/>) (automatic generation from text input)

## ➤ Maps by coding

- Some folks prefer using code to generate maps. It can be faster if precise layout is not an issue. See for example:
- **Graphviz**





<https://www.youtube.com/watch?v=pMMRE4Q2FGk>  
<https://www.youtube.com/watch?v=HxrbptT3hUo>  
<https://www.youtube.com/watch?v=rkgUP2lgqs4>  
<https://www.youtube.com/watch?v=M3aZNaPY88Y>  
<https://www.youtube.com/watch?v=IkH0EGYqWO0>  
<https://www.youtube.com/watch?v=1fEC5rFCATw>  
<https://www.youtube.com/watch?v=AEKJ1nHxH1w>  
<https://www.youtube.com/watch?v=vGyjF9Ngd8Y>  
<https://www.youtube.com/watch?v=r1aYWbLj0U8>  
<https://www.youtube.com/watch?v=0pPriKf1f2o>  
<https://www.youtube.com/watch?v=jC4Vaf2MrN0>  
<https://www.youtube.com/watch?v=JJu1uFokP6I>  
<https://www.youtube.com/watch?v=bQlgx5biqCQ>  
<https://www.youtube.com/watch?v=9Pzz6rDMd8c>  
<https://www.youtube.com/watch?v=mHdy1xS59xA>





# THANK YOU

